Integrable vertex models and Symmetric functions

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Lecture 1

Sym. functions

Schur poly $s_\lambda(x)$, $\lambda = (\lambda_1, \lambda_2, \dots \lambda_\ell)$ partition

$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_\ell \geq 0$

Set of variables $x = (x_1, \dots x_n)$

Def

$$s_\lambda(x) = \frac{\det\left(x_i^{\lambda_j + n - j}\right)_{1 \leq i,j \leq n}}{\det\left(x_i^{n-j}\right)_{1 \leq i,j \leq n}}$$

where $\lambda_j = 0$ if $j > \ell$

if $n \geq \ell$

Set $s_\lambda(x) = 0$ if $n < \ell$.

Denominator is $\left(\prod_{1 \leq i < j \leq n} (x_i - x_j)\right) \prod_{i=1}^{n} x_i^{\square}$ (Vandermonde)

Called (bi)alternant formula, Weyl character formula

Schur poly. ubiquitous
- → Rep thry
- → Alg Geom
- → Prob. theory

① Combinatorial realization

$$s_\lambda(x) = \sum_{T \in SSYT_n(\lambda)} x^T = \sum_{\text{"vertex model of } \lambda\text{"}} \text{Weight}(VM)$$

T ranges over all semistandard Young tableaux of λ w/ fillings in $\{1, 2, \dots, n\}$

Young diagram of λ $(n=6)$

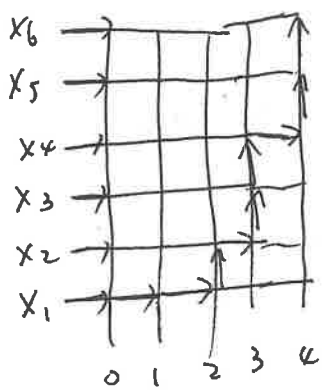
1	1	2	4
2	3	4	6
5	5	5	

strictly $\uparrow$ columns

weakly $\uparrow$ rows

$$X^T = \prod_{i=1}^{n} X_i^{\#\, i \in T} = x_1^2 x_2^2 x_3 x_4^2 x_5^3 x_6$$

Take a rectangle of height n and width $\lambda_1 + 1$



First row of SSYT governs bottommost path

#s in cells say which rows path turns right

Simplest case of a vertex

weight each horizontal arrow in row by x_i

② <u>Branching identity</u>

$$\underset{(x_1,\dots,x_n)}{S_\lambda(x)} = \sum_{\mu < \lambda} x_n^{|\lambda| - |\mu|} S_\mu(x_1, x_2, \dots, x_{n-1})$$

μ interlaces λ

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \dots \geq \mu_{\ell-1} \geq \lambda_\ell$$

$$|\lambda| = \sum_{i=1}^{\ell} \lambda_i$$

③ <u>Symmetry</u>.

$$s_\lambda(x_1, x_2, \dots, x_i, x_{i+1}, \dots, x_n)$$

$$= s_\lambda(x_1, x_2, \dots, x_{i-1}, x_{i+1}, x_i, \dots, x_n)$$

④ <u>Symmetization formula</u>

$$s_\lambda(x) = \sum_{\sigma \in S_n} \sigma\left(x_1^{\lambda_1} x_2^{\lambda_2} \cdots x_n^{\lambda_n} \prod_{1 \le i < j \le n} \frac{x_i}{x_i - x_j} \right)$$

$$\sigma f(x_1, x_2, \dots, x_n) = f(x_{\sigma(1)}, x_{\sigma(2)}, \dots, x_{\sigma(n)})$$

⑤ <u>Cauchy identity</u> $y = (y_1, y_2, \dots, y_m)$

Assume $|x_i y_j| < 1$, $\forall i, j$.

$$\sum_\lambda s_\lambda(x)\, s_\lambda(y) = \prod_{\substack{1 \le i \le n \\ 1 \le j \le m}} (1 - x_i y_j)^{-1}$$

⑥ <u>Structure constants</u>

For any λ, μ, ν, exists $c_{\lambda,\mu}^{\nu} \in \mathbb{Z}$ s.t.

$$s_\lambda(x)\, s_\mu(x) = \sum_\nu c_{\lambda\mu}^{\nu}\, s_\nu(x)$$

These Littlewood–Richardson coefficients have <u>combinatorial realizations</u> (puzzles)

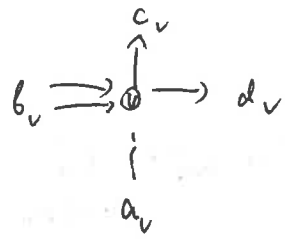
↳ more involved than ①

<u>Goal</u> Explain how ①-⑥ can be proven for Schur polynomials and for much more general sym. polynomials through integrable vertex models.

Vertex Models (Simplest case)

Weighted up-right paths on a rectangle R in $\mathbb{Z}^2$.

At each vertex, arrows (paths) entering and exiting



$a_v = 0$
$b_v = 2$
$c_v = 1$
$d_v = 1$

Call (a_v, b_v, c_v, d_v) arrow configuration at v

Arrow-conservation
$a_v + b_v = c_v + d_v$

For each $v \in R$, assign a variable called a spectral parameter $z_v \in \mathbb{C}$

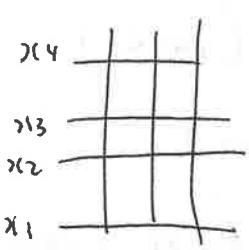
Fix a weight function $w_z : \mathbb{Z}^4_{\geq 0} \longrightarrow \mathbb{C}$

$$\text{Weight (Configuration)} = \prod_{v \in R} w_{z_v}(a_v, b_v, c_v, d_v)$$

Partition function
$$Z = \sum_{\substack{\text{configurations w/} \\ \text{fixed entrance/exit data}}} \text{Weight (Configuration)}$$

Gibbs measure

Example $w_z(a,b,c,d) = \mathbb{1}_{a+b=c+d} \, \mathbb{1}_{b,d \in \{0,1\}} \, z^d$



Put $z_v = x_i$ if v is in row i

Then $s_\lambda(x)$ = partition func. of vertex model w/ λ dictating top exit, "Full" left entrance. Nothing on B/R.

SSYT $\subsetneq$ Vertex models

$\rightsquigarrow$ vertex models and their partition functions describe an extremely broad class of objects

<u>Issue</u>. does not directly give a means to study them.

<u>Integrability</u>. Impose two conditions. 1) on z_v ("Rapidity setup")

2) On w ("Yang-Baxter equation")

1) For each horizontal line ℓ, assign a rapidity parameter x_ℓ and for each vertical line ℓ' assign $y_{\ell'}$.

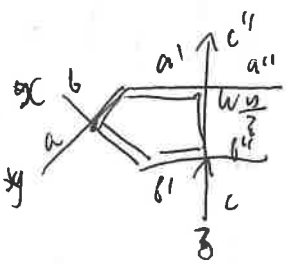
Assume $z_v = \dfrac{x_\ell}{y_{\ell'}}$ where $v = \ell \cap \ell'$ (in literature, sometimes $z_v = \dfrac{y_{\ell'}}{x_\ell}$)

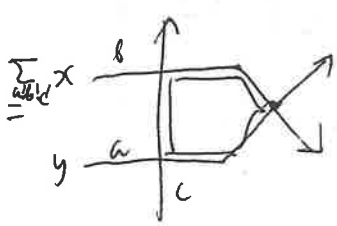
2) Yang-Baxter equation

Source of "solvability"/"integrability"

<u>Assume</u> exists weight function $r_z : \mathbb{Z}_{\geq 0}^4 \to \mathbb{C}$ denoted

s.t.

$\sum_{a'b'c'}$ 

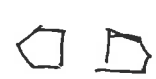
$\sum_{a'b'c'}$ 

Fix boundary data

(a, b, c, a'', b'', c'')

Fix rapidities (x, y, z)

$\rightsquigarrow$ fix spatial parameter

Sum over trajectories of paths on 

Explicitly,

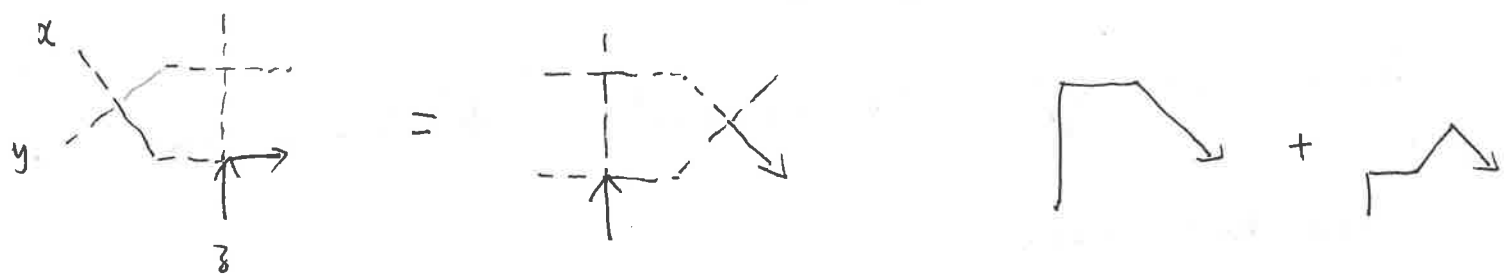
$$\sum_{a',b',c'} r_{\frac{x}{y}}(a,b,a',b')\; w_{\frac{x}{z}}(c,b',c',b'')\; w_{\frac{y}{z}}(c',b',c'',b'')$$

$$= \sum_{a',b',c'} w_{\frac{y}{z}}(c,a,c',a')\; w_{\frac{x}{z}}(c',b,c'',b')\; r_{\frac{x}{z}}(a',b',a'',b'')$$

<u>Example</u>. $w_z(a,b,c,d) = \mathbb{1}_{a+b=c+d}\; \mathbb{1}_{b,d\in\{0,1\}}\; z^d$

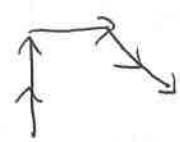
r_z

$1 \quad 1 \quad 1-z^{-1} \quad 1 \quad z^{-1}$



$w_{\frac{x}{z}}\Big($ $\Big) = \frac{x}{z}$

$(1,0,0,1)$



$w_{\frac{y}{z}}\Big($ $\Big) = \frac{y}{z}$

$r_{\frac{x}{y}}\Big($ $\Big) = 1$

$w_{\frac{y}{z}}\Big($ $\Big) = 1$

$w_{\frac{x}{z}}\Big($ $\Big) = \frac{x}{z}$

$r_{\frac{x}{y}}\Big($ $\Big) = 1 - \frac{y}{x}$

$\frac{x}{z}\left(1-\frac{y}{x}\right) = \frac{x}{z} - \frac{y}{z}$

① How to come up w/ (w, r) ?

(Fusion)

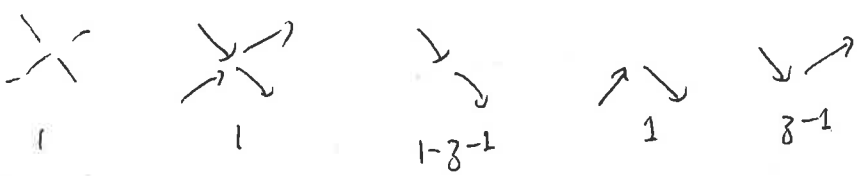
② Which sym. function = partition function of integrable vertex models?

③ How to use integrability to prove properties of functions ?

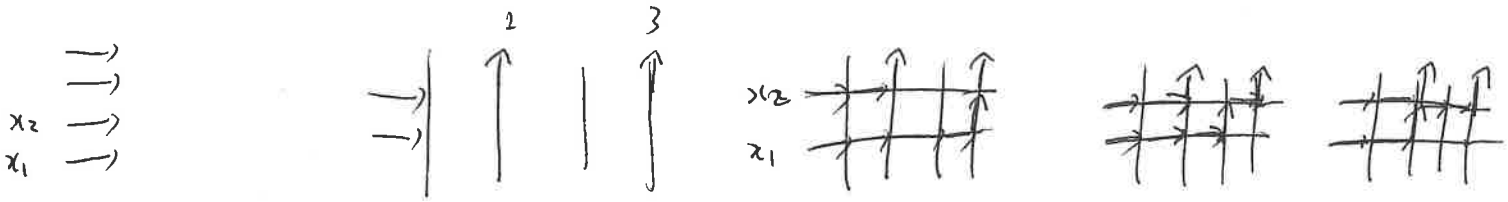
Lecture 2

(Bosonic Hall-Littlewood type)

$$W_z(a,b,c,d) = W_z\Big(b \rightrightarrows \overset{c}{\underset{a}{\uparrow\uparrow}} \rightrightarrows d\Big) = z^d \, \mathbb{1}_{b,d\in\{0,1\}} \, \mathbb{1}_{a+b=c+d}$$



① Combinatorial realization $\Leftrightarrow$ vertex model



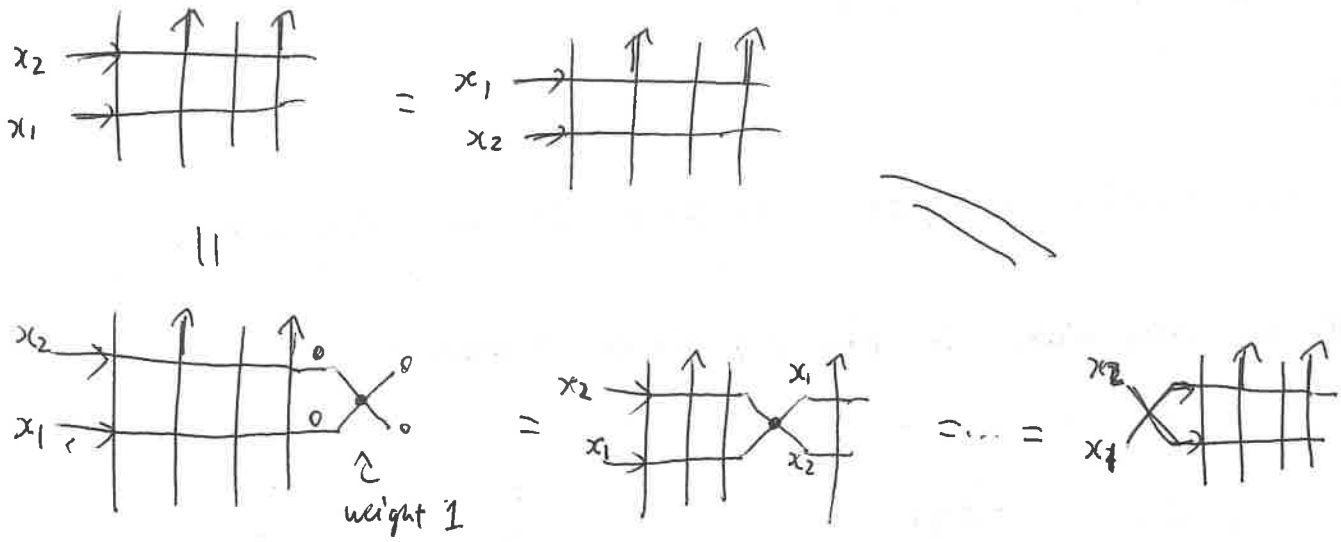
$$x_1^3 x_2 + x_1^2 x_2^2 + x_1 x_2^3$$

$$= s_{(3,1)}(x_1, x_2)$$

② Branching "Peeling off" the top row of the vertex model.

③ Symmetry. $\sigma_i\big(s_\lambda(x_1,\dots,x_n)\big) = s_\lambda(x_1,\dots,x_n)$

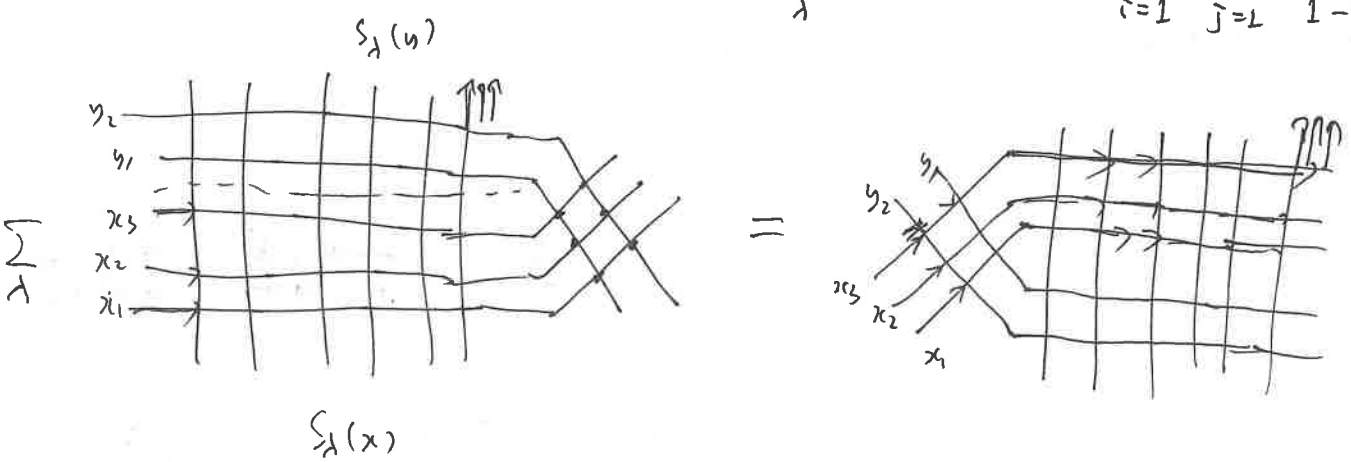
$\sigma_i = (i, i+1)$



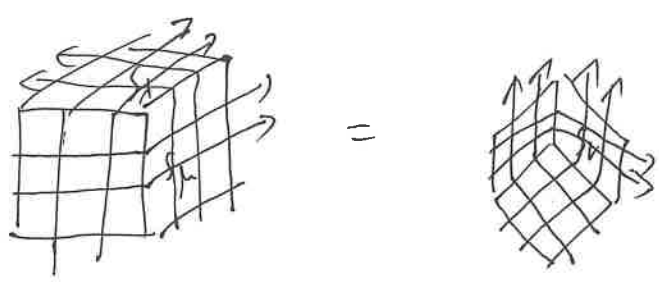
④ <u>Symmetrization formula</u> $\quad S_\lambda(x) = \sum_{\sigma \in \mathfrak{S}_n} \sigma\left(x_1^{\lambda_1} x_2^{\lambda_2} \cdots x_n^{\lambda_n} \prod_{i<j} \frac{x_i}{x_i - x_j} \right)$

Algebraic Bethe ansatz

⑤ <u>Cauchy identity</u> $\quad |x_i y_j| < 1 \rightsquigarrow \sum_\lambda S_\lambda(x) S_\lambda(y) = \prod_{i=1}^{n} \prod_{j=1}^{m} \frac{1}{1 - x_i y_j}$



⑥ Littlewood-Richardson coefficients



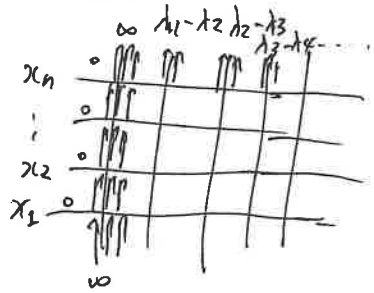
Use "colored" or "higher rank" vertex models.

$$S_\lambda S_\mu = \sum_\nu \underbrace{c_{\lambda\mu}^{\nu}}_{\text{LR coeff.}} S_\nu$$

$$W_z(a,b,c,d) = \mathbb{1}_{a+b=c+d}\ \mathbb{1}_{a\geq d}\ z^d$$

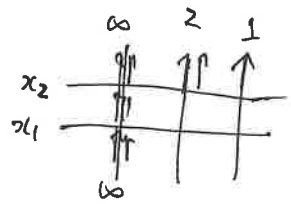
(Bosonic q-Whittaker type)

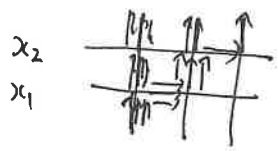
$x = (x_1, x_2, \dots, x_n)$

$S_\lambda(x) =$ 

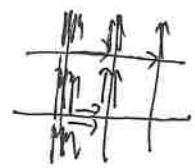
e.g. $\lambda = (3,1)$

$n=2$

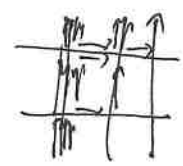




$x_1^3 x_2$



$x_1^2 x_2^2$



$x_1 x_2^3$

(Fermionic vertex model)

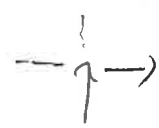
$U_z(a,b,c,d)$ Require $a+b=c+d$, $a,b,c,d \in \{0,1\}$

1

1

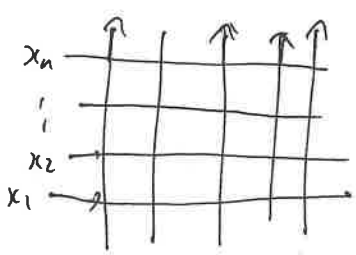
z

1



1

arrows exit at sites $\lambda_i + n - i$

$S_\lambda(x_1, x_2, \dots, x_n) =$ 

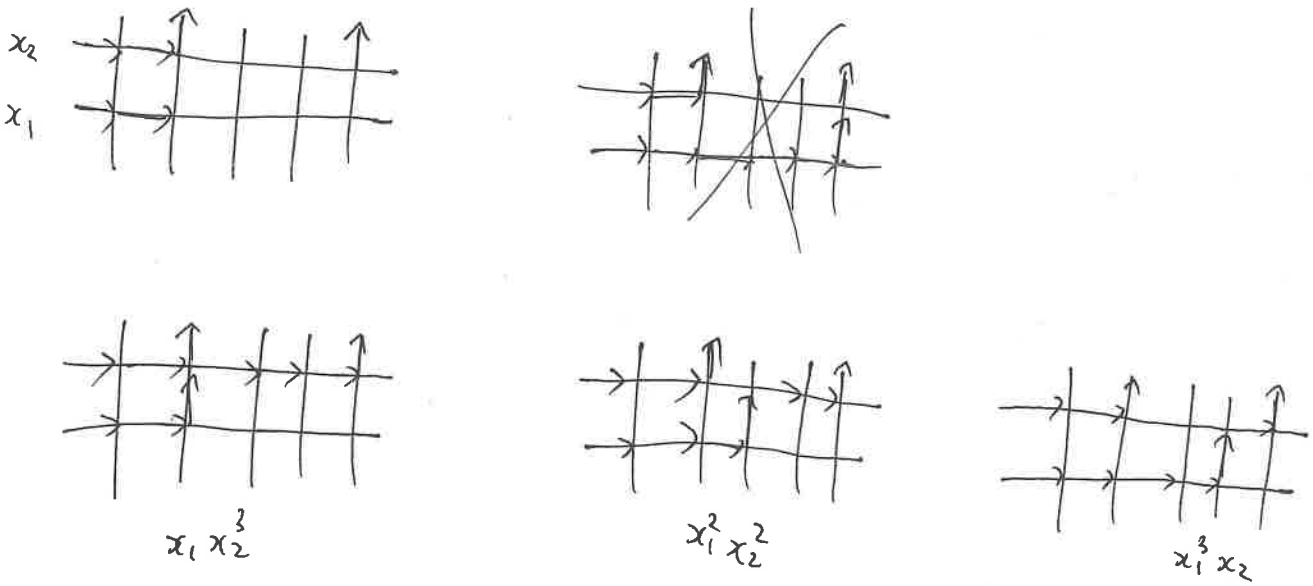
Bottommost past exists at $\lambda_1 + n - 1$

second bottommost $\lambda_2 + n - 2$

$\vdots$

λ_n

$\lambda = (3,1)$, $n=2$



Parameters entering weights

① Rapidities: Horizontal ✓
Vertical ?

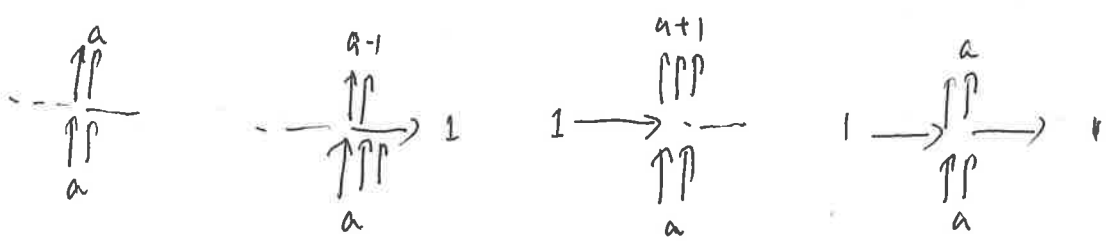
② Quantization parameter q (fixed across the vertex model) ($q=0$ above)

③ Spin variables s. Also associated w/ vertical and horizontal lines (just as rapidities)

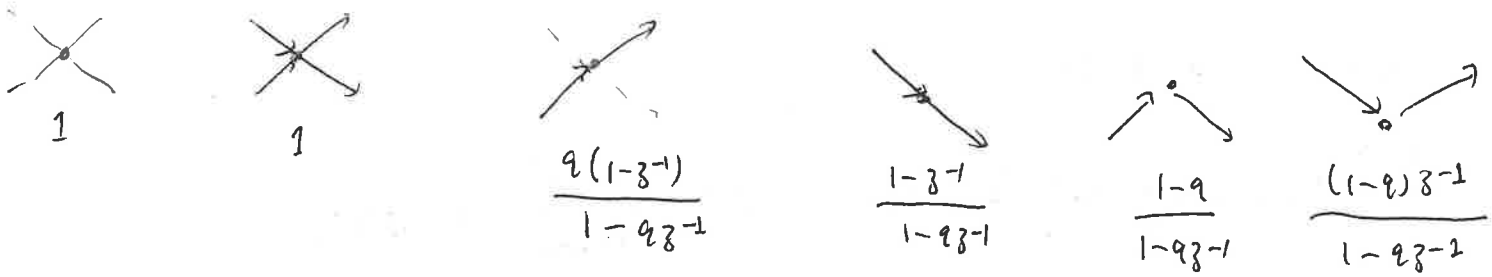
④ Colors of arrows (Only had one arrow colour) (above $s=0$)

<u>Bosonic Hall-Littlewood</u> = ① + Quantization par. q

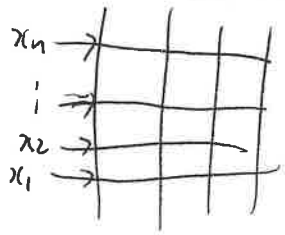
Have the property that $b, d \in \{0,1\}$.



$1 \qquad 3 \qquad 1-q^{a+1} \qquad 3$



$P_\lambda(x_1, \cdots x_n) =$



arrows exist at sites (λ_i)

$P_\lambda(x_1, \cdots, x_n)$ is the Hall-Littlewood polynomial.

$$P_\lambda(x_1, \cdots, x_n) = (1-q)^n \sum_{\sigma \in S_n} \sigma\left(x_1^{\lambda_1} x_2^{\lambda_2} \cdots x_n^{\lambda_n} \prod_{i<j} \frac{x_i - q x_j}{x_i - x_j}\right)$$

(symmetization formula for HL)

<u>Bosonic q-Whittaker</u> ② + quantization par. q

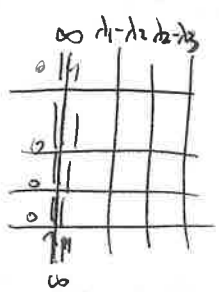
$$W_z^q(a, b, c, d) = \mathbb{1}_{a+b=c+d} \, \mathbb{1}_{a \geq d} \, z^d \frac{(q;q)_c}{(q;q)_d (q;q)_{a-d}}$$

$$(q;q)_k = (1-q)(1-q^2)\cdots(1-q^k)$$

When $a = c = \infty$,

$$W_z^q(\infty, b, \infty, d) = \frac{z^d}{(q;q)_d}$$

$\mathbb{F}_\lambda^q(x_1, \cdots, x_n) =$

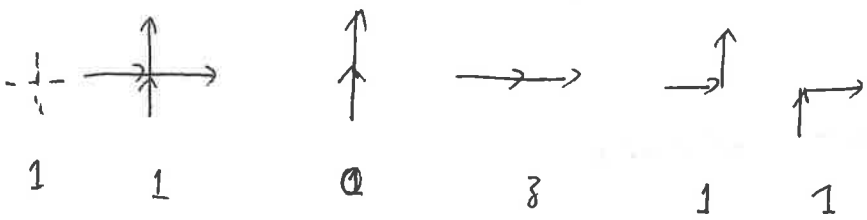


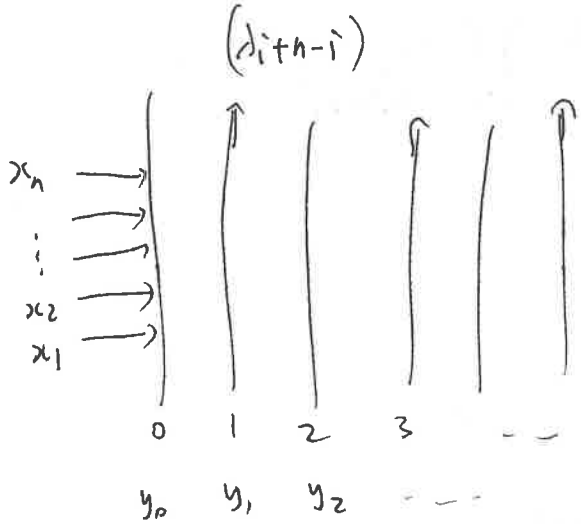
$=$ q-Whittaker polynomial

Bosonic HL weights & Bosonic q-Whittaker weights are both special cases of a more general family.

$$\sum_{\lambda} P_\lambda(x_1, x_2, \dots x_n)\, \mathbb{F}_\lambda(y_1, y_2, \dots, y_m) = \prod_{\substack{1\le i\le n \\ 1\le j\le m}} (1 + x_i y_j)$$

<u>Lecture 3</u> Fermionic vertex model for Schur





$= s_\lambda(x_1, x_2, \dots, x_n)$

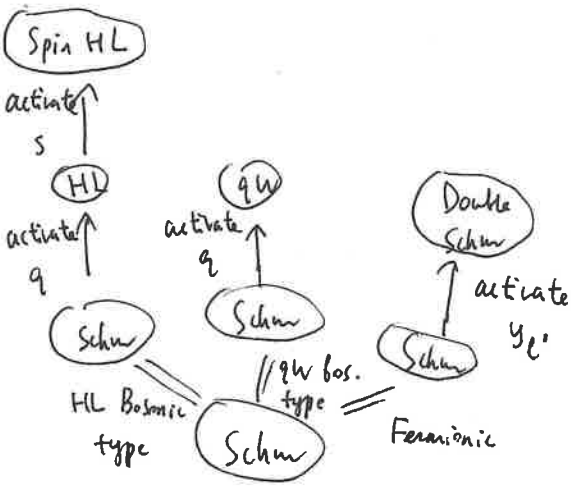
Now set $z_{ij} = x_j + y_i$

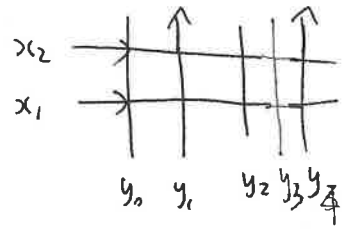
at vertex $(i,j) \in \mathbb{Z}^2$

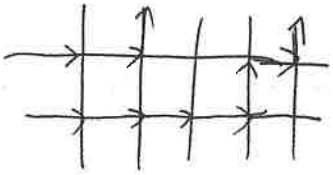
Now, partition function $= s_\lambda(x \| y)$ $(y_0, y_1, \dots)$

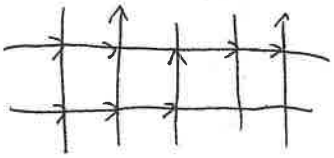
$(x_1, x_2, \dots, x_n)$

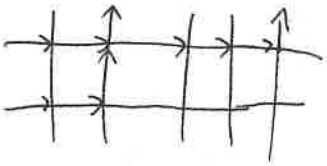
"Factorial Schur" / "Double Schur" / "Interpolation Schur"











$$(x_2+y_0)(x_1+y_0)(x_1+y_1)(x_1+y_2)$$
$$+ \cdots$$
$$+ \cdots$$

Observe that the partition function vanishes at $x_1 = -y_0$ and $x_2 = -y_0$

<u>Vague Statement</u>: There exists a family of vanishing points $x_i = y_{\sigma_i}$ that uniquely characterizes $s_\lambda(x \| y)$.

<u>Bi-Alternant formula</u>

$$s_\lambda(x_1, x_2, \cdots, x_n \| y) = \frac{\det\left[(x_i|y)^{\lambda_j + n - j}\right]}{\prod_{i<j}(x_i - x_j)} \leftarrow \det\left(x_i^{n-j}\right)_{i,j}$$

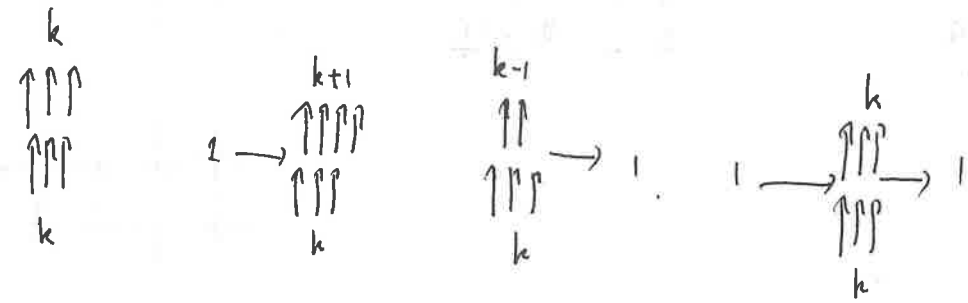
$$(x|y)^k = (x+y_0)(x+y_1)\cdots(x+y_{k-1})$$

<u>Tableaux formula</u> $\displaystyle\sum_{T \in SSYT_n(\lambda)} (x|y)^T = s_\lambda(x\|y)$ where $(x|y)^T = \prod_{(i,j)\in T}\left(x_{T(i,j)} + y_{T(i,j)+j-i-1}\right)$

Activating s

Spin HL

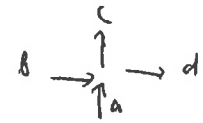
Bosonic HL weights



Last time $\quad 1 \qquad 1-q^{k+1} \qquad x \qquad x$

$$\frac{1-sq^k x}{1-sx} \qquad \frac{1-q^{k+1}}{1-sx} \qquad \frac{(1-s^2q^{k-1})x}{1-sx} \qquad \frac{x-sq^k}{1-sx}$$

$s=0$

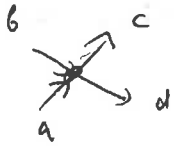


Call these weights $w_x(a,b,c,d)$

(Spin HL weights)

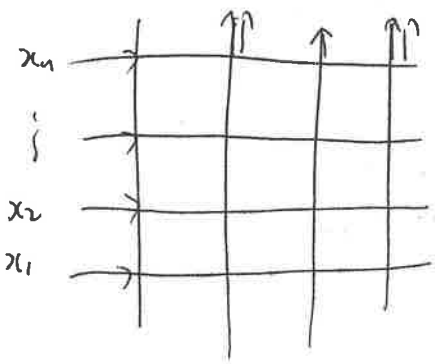
Either fix s across the lattice (as for q), or can have s depend on columns ($s = s_{\ell'}$)

Last time we gave 2 weights



<u>Claim</u> Same 2-weights satisfy YBE w.r.t.

Exit at (λ_i)



$= F_\lambda^{s,q}(x_1, x_2, \dots, x_n)$ Spin Hall-Littlewood polynomials

<u>Symmetrization formula</u>

$$F_\lambda(x) = \frac{(1-q)^n}{\prod_{i=1}^n (1-sx_i)} \sum_{\sigma \in \mathfrak{S}_n} \sigma\left(\prod_{i<j} \frac{x_i - qx_j}{x_i - x_j} \prod_{i=1}^n \left(\frac{x_i - s}{1-sx_i}\right)^{\lambda_i} \right)$$

Stochastic weights

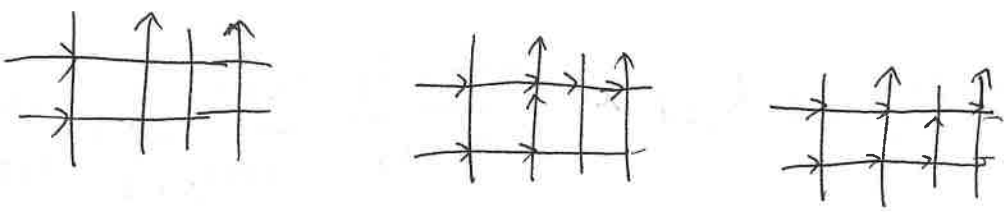
only changes F up to a factored quantity.

$$L_x^{\text{Stoch}}(a,b,c,d) = (-s)^d \boxed{\frac{(s^2;q)_c}{(s^2;q)_a} \cdot \frac{(q;q)_a}{(q;q)_c}} \, w_x(a,b,c,d)$$

Recall $(a;q)_k = (1-a)(1-aq)\dots(1-aq^{k-1})$

"Gauge transformation" / "Conjugation" / "Drinfeld twist"

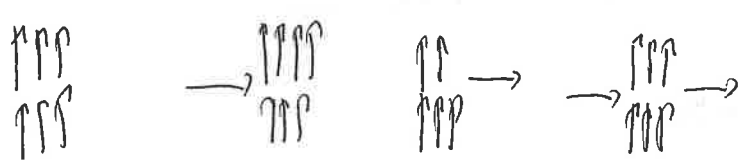
Consider a transformation of weights $w(a,b,c,d) \mapsto \frac{f(c)}{f(a)} w(a,b,c,d)$



Both configurations have weight under w and under $\tilde{w}$

$$\text{Ratio} = \frac{\text{weight under } w}{\text{weight under } \tilde{w}} \quad \text{does not depend on configuration.}$$

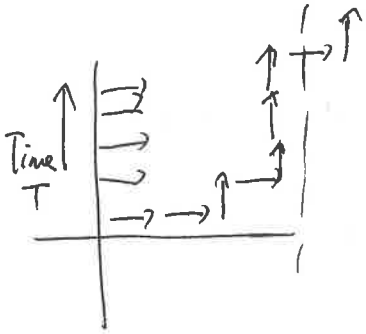
$$= \prod_{\text{columns}} \frac{f(\text{Exit})}{f(\text{Entrance})}$$



$$\frac{1-sq^k x}{1-sx} \qquad \frac{1-s^2q^k}{1-sx} \qquad \frac{sq^k x - sx}{1-sx} \qquad \frac{s^2q^{k-1} - sx}{1-sx}$$

Fix the incoming, sum to 1 over outgoing possibilities

$\to$ Markov process



<u>Question</u>. How many arrows cross the line $\{x = s\}$ after time T?

<u>Fact</u> $$\frac{N(s,T) - \mathbb{E}[N(s,T)]}{T^{1/3}} \longrightarrow TW_2$$

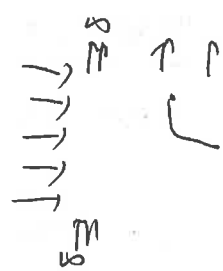
Set $s = q^{-1/2}$ $\rightsquigarrow$ $1 - s^2 q^k = 0$ if $k = 1$

<u>Spin q-Whittaker</u>

$$w\left(b \to \overset{c}{\uparrow} \to d \atop a \right) = \mathbb{1}_{a+b=c+d}\, \mathbb{1}_{a \geq d}\, x^d \frac{\left(\frac{-s}{x}; q\right)_d}{(q;q)_d} \frac{(-sx; q)_{a-d}}{(q;q)_{a-d}} \frac{(q;q)_c}{(s^2;q)_c}$$

(Setting $s=0$ recovers qW bosonic weights)

Same vertex model as for qW.



$(\lambda_i - \lambda_{i+1}) = \mathbb{F}^{s,q}_\lambda(x_1, \dots, x_n)$

(Spin q-Whittaker polynomials)

These $\mathbb{F}^{s;q}_\lambda(x_1, \dots, x_n)$ satisfies very similar properties to qW polys.

Just as $\sum_\lambda (HL_\lambda)(qW_\lambda) =$ Factored

(Cauchy identity between HL and qW)

$$\sum_\lambda (sHL_\lambda)(sqW_\lambda) = \text{Factored}$$

$$\mathbb{1}^{\text{Stoch}}_{x,s}(a,b,c,d) = \mathbb{1}_{a+b=c+d}\,\mathbb{1}_{a\geq d}\,(-sx)^d\,\frac{\left(-\frac{s}{x},q\right)_d\,(-sx;q)_{a-d}}{(s^2;q)_a}\,\frac{(q;q)_a}{(q;q)_d\,(q;q)_{a-d}}$$

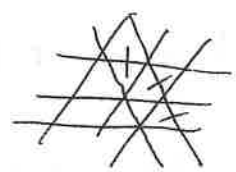
<u>Fact</u>. $\sum_{c,d} \mathbb{1}^{\text{Stoch}}_{x,s}(a,b,c,d) = 1$ for fixed a, b.

If $q \to 1$ carefully, get random polymers (version of random walks in random environments)

Take first vertex model for Schur.

$$w\left(b \to \begin{smallmatrix} c \\ \uparrow \\ \uparrow \\ a \end{smallmatrix} \to d\right) = x^d\,\mathbb{1}_{a+b=c+d}\,\mathbb{1}_{b,d\in\{0,1\}}$$

Lozenge tiling: tiling of the triangle lattice by rhombi



<u>Lecture 4</u> <u>Today and next time</u>. Where these weights come from (fusion)

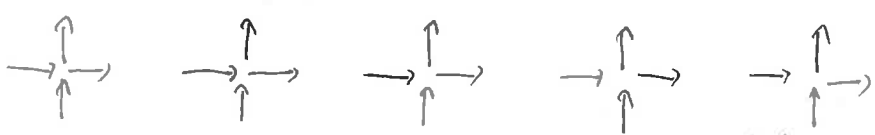
<u>Basic idea</u> (1) One family (simple) of weights satisfying YBE

(2) Create more solutions from this family by "projection".

<u>Colored six vertex model</u>

Fix $n \geq 1$

At each vertex, arrows enter and exit, but now they have a label (= colour) $i \in [0,n]$



— colour i z spectral par.

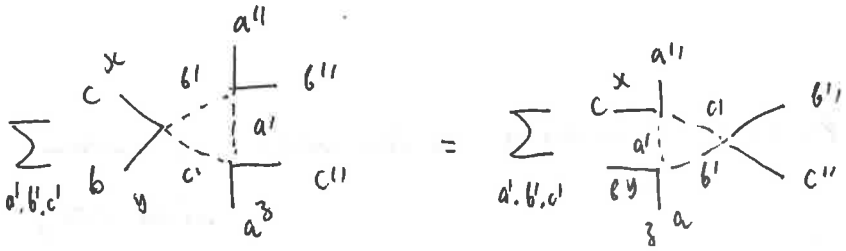
— colour j $\uparrow$ (depend. on vertex)

1 $\quad \frac{q(1-z)}{1-qz}$ $\quad \frac{1-z}{1-qz}$ $\quad \frac{1-q}{1-qz}$ $\quad \frac{z(1-q)}{1-qz}$ $\quad R_z(a,b,c,d)$ $\in[0,n]$ $i<j$

ratio of $\frac{y_{e'}}{x_e}$

Fact 1 For fixed $a, b \in [0,n]$, $\sum_{c,d \in [0,n]} R(a,b,c,d) = 1$ (stochasticity)

Theorem These weights satisfy YBE



Proof. Direct check.

Last time, we have W satisfying YBE w.r.t. r $\times$
$+$

Now We have r $+$ satisfying YBE w.r.t. r $\times$.

(First) fusion turns $r \rightsquigarrow w$.

Second fusion (not today) r $\times$ $\rightsquigarrow$ $\tilde{w}$.

Typical to organize these weights into a $(n+1)^2 \times (n+1)^2$ matrix

indexing lines crossing to form the vertex

$R_z^{12} = R^{12}(z) \longrightarrow$ Think of this matrix as acting on $V_1 \otimes V_2 \to \mathbb{C}^{n+1}$, basis labeled by $\{e_0, e_1, \dots, e_n\}$

spectral par.

Entries are $R_z^{12}(a, b, c, d) = R_z(a,b,c,d)$ in $[0,n]$

$$R_z^{12} = \sum_{a,b,c,d} R_z(a,b,c,d)\, E^1_{ac} \otimes E^2_{bd}$$

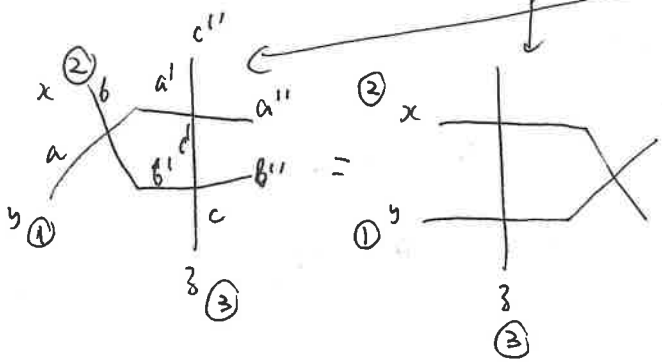
$E^\ell_{ij} = (n+1) \times (n+1)$ mat. w/ 1 in (i,j) entry and 0 elsewhere

Example $n=1$.

$$R_z^{12} = \begin{array}{c} (c,d) \to \\ \\ \\ \\ \end{array} \begin{array}{c} \begin{array}{cccc} (0,0) & (0,1) & (1,0) & (1,1) \end{array} \\ \begin{bmatrix} 1 & & & \\ & \frac{1-z}{1-qz} & \frac{(1-q)z}{1-qz} & \\ & \frac{1-q}{1-qz} & \frac{q(1-z)}{1-qz} & \\ & & & 1 \end{bmatrix} \end{array} \begin{array}{c} (a,b) \\ \downarrow \\ (0,0) \\ (0,1) \\ (1,0) \\ (1,1) \end{array}$$

YBE.

$$R^{12}\left(\frac{y}{x}\right) R^{32}\left(\frac{z}{x}\right) R^{13}\left(\frac{z}{y}\right) = R^{13}\left(\frac{z}{y}\right) R^{32}\left(\frac{z}{x}\right) R^{12}\left(\frac{y}{x}\right)$$



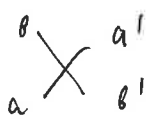
$R^{ij}(z)$ acts on $V_i \otimes V_j$

equality as elements of

$\mathrm{End}(V_1 \otimes V_2 \otimes V_3)$

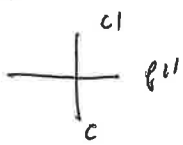
$$R^{13}\left(\frac{z}{y}\right) R^{32}\left(\frac{z}{x}\right) R^{12}\left(\frac{y}{x}\right) e_a \otimes e_b \otimes e_c$$

Sum over



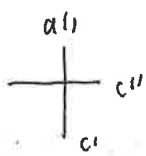
$e_{a'} \otimes e_{b'}$

Sum over



$e_{c'} \otimes e_{b''}$

Sum over a'



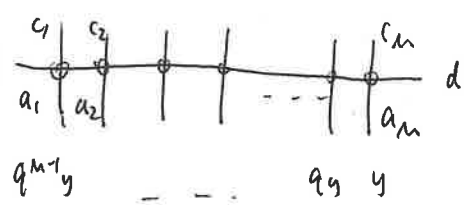
$e_{a''} \otimes e_{c''}$

Fix $M \geq 1$.

<u>Fusion</u>. Define

"row vertex" $R_{\frac{y}{x}}\left((a_1, a_2, \cdots a_M), b, (c_1, c_2, \cdots, c_M), d\right)$

$=$



Rapidity x

Element of $\mathbb{Z}^{n+1}_{\geq 0}$ $(A_0, A_1, \cdots, A_M)$

Think of $A_i = \#$ column i arrows entering from bottom

Define $L_{\frac{y}{x}}\left(A, b, (c_1, c_2, \cdots c_M), d\right)$

$$= \sum_{a_1, a_2, \cdots, a_M} q^{\mathrm{inv}(a_1, a_2, \cdots, a_M)} R_{\frac{y}{x}}\left((a_1, a_2, \cdots, a_M), b, (c_1, c_2, \cdots, c_M), d\right)$$

s.t. $\#(a_i = k) = A_k$ $\quad \mathrm{inv}(\vec{a}) = \#\{a_i > a_j, i < j\}$

Proposition $q\text{-inv}(c_1, c_2, \cdots, c_M)$ $L_{\frac{y}{x}}(A, b, (c_1, c_2, \cdots, c_M), d)$ does not depend on $(c_1, c_2, \cdots, c_M)$ with given content:

$$C = (C_0, C_1, \cdots, C_n)$$

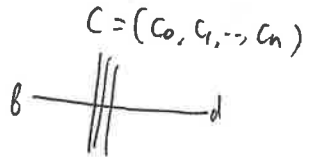
where $C_i = \#(c_k = i)$

Def. Set $L_{\frac{y}{x}}(A, b, C, d) = (*)$,

$b, d \in \{0, 1, 2, \cdots, n\}$ index colours

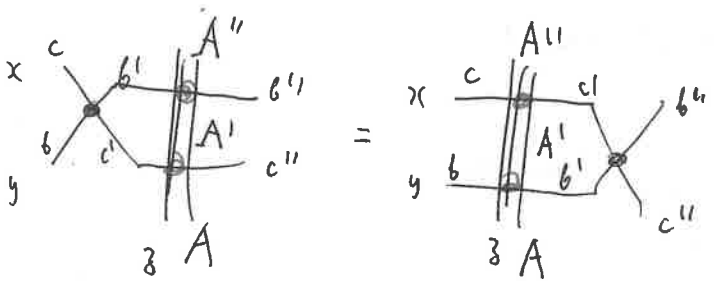
$A, C \in \mathbb{Z}_{\geq 0}^{n+1}$ index contents

$L_{\frac{y}{x}}(A, b, C, d) =$



A_i, C_i tell how many arrows of colour i enter / exit vertically

Prop. L satisfies YBE w.r.t. R



- R-weight
- L-weight

Rmks 1) We can compute L-weights (next week)

2) $n=1$, $s = q^{-\frac{M}{2}}$ $\longrightarrow$ Spin HL wts

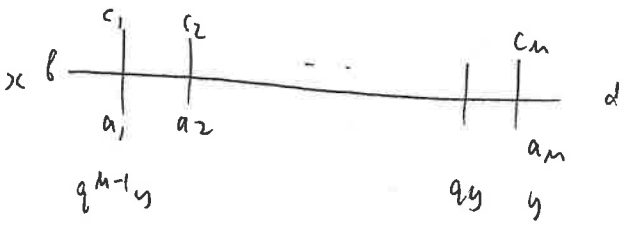
3) weights will be rat'l in q^M.

Lecture 5

Row vertex

$$R_{\frac{y}{x}}\Big(\overset{(a_k)}{(a_1, a_2, \cdots, a_M)}, b, \overset{(c_k)}{(c_1, c_2, \cdots, c_M)}, d\Big)$$

$=$



$$L_{\frac{y}{x}}\Big(\overset{A:\,(A_0, A_1, \cdots, A_n)}{A}, b, (c_k), d\Big)$$

$$= \sum_{(a_k)} q^{\mathrm{inv}(a_k)} R_{\frac{y}{x}}\big((a_k), b; (c_k), d\big)$$

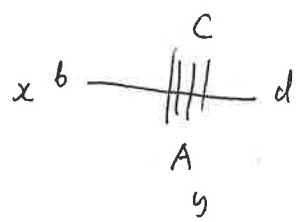
$\#\{k : a_k = j\} = A_j$

Prop 1 q-Exchangeability

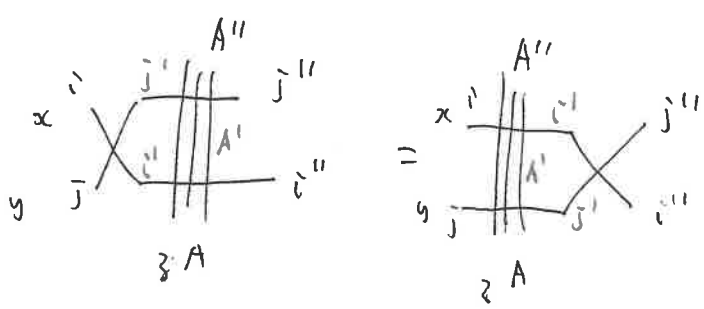
(*) $L_{\frac{y}{x}}(A, b; (c_k), d)\, q^{-\mathrm{inv}(c_k)}$ does not depend on (c_k) w/ fixed content

$C = (C_0, C_1, \cdots, C_n)$, $\quad C_j = \#\{k : c_k = j\}$

Define $L_{\frac{y}{x}}(A, b; C, d) = (*)$



Prop 2 (YBE)

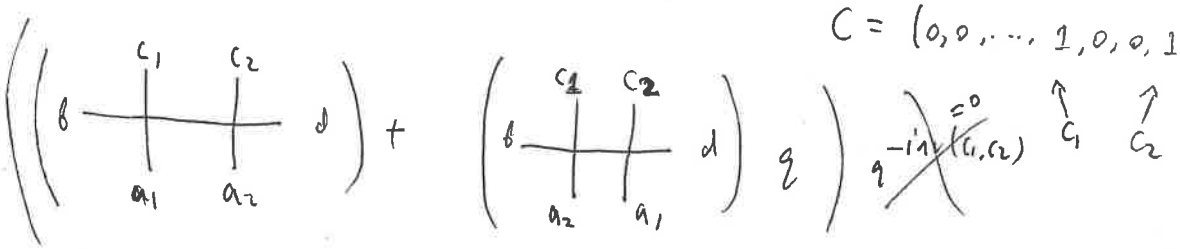


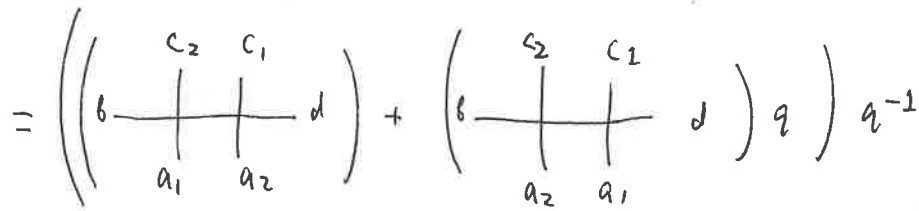
sum over (A', i', j') on both sides.

Proof of prop 1 $\quad M = 2.$

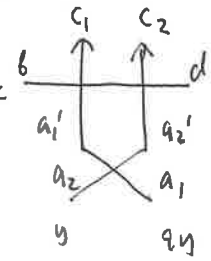
Example $\quad a_1 < a_2, \quad c_1 < c_2 \quad \longrightarrow \quad A = (0, 0, \dots, \overset{a_1}{1}, 0, 0, \dots, 0, \overset{a_2}{1}, 0, 0, \dots, 0)$

$C = (0, 0, \dots, \underset{c_1}{1}, 0, 0, \underset{c_2}{1}, 0, 0, \dots, 0)$



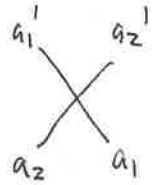


Fact. $\quad L_{\frac{y}{x}}(A, b, (c_k), d) = x$



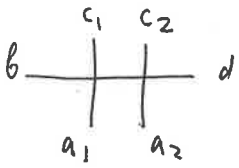
$(q+1)^{\mathbb{1}_{a_1 \neq a_2}} \qquad (**)$

Proof $\quad$ If $a_1 = a_2$, $\quad a_1' = a_2' = a_1 = a_2 \quad \longrightarrow$



$= 1$

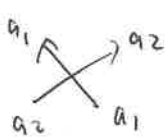
$\longrightarrow \quad (**) =$

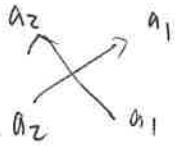


✓

$a_1 \neq a_2$.

If $a_1 < a_2$,



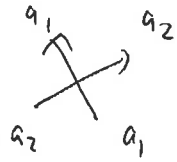


Here $z = q$

$$\frac{1-q}{1-q^2} = \frac{1}{1+q} \qquad \frac{q(1-q)}{1-q^2} = \frac{q}{1+q}$$

$$= \frac{q^{inv(a_1', a_2')}}{1+q} \qquad = \frac{q^{inv(a_1', a_2')}}{1+q}$$

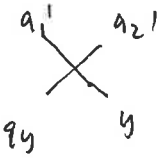
If $a_1 > a_2$,



$$= \frac{q}{1+q} = \frac{q^{\mathrm{inv}(a_1', a_2')}}{1+q}$$

$$= \frac{1}{1+q} = \frac{q^{\mathrm{inv}(a_1', a_2')}}{1+q}$$

<u>Projector property</u>

Output of 

only depends on contents of input.

$R(q)$ $(n+1) \times (n+1)$ matrix

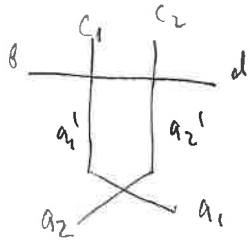
$$R(q)\, e_i \otimes e_i = e_i \otimes e_i$$

$$R(q)\, e_i \otimes e_j = R(q)\, e_j \otimes e_i = \frac{1}{1+q} e_i \otimes e_j + \frac{q}{1+q} e_j \otimes e_i \qquad (i < j)$$

Consequence. $\mathrm{Im}\, R(q) = \mathrm{Span}\{ e_i \otimes e_i,\ e_i \otimes e_j + q\, e_j \otimes e_i \}$

$= $ "q- Symmetric tensors"

<u>End of proof of fact</u>

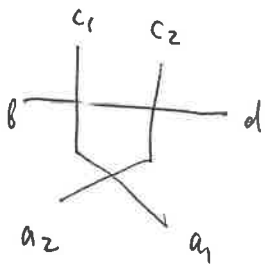


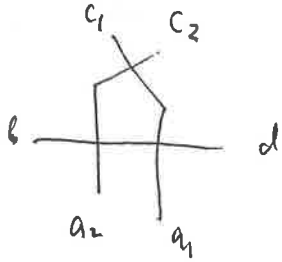
$$(q+1) = (q+1) \sum_{(a_1', a_2')} \frac{q^{\mathrm{inv}(a_1', a_2')}}{1+q} \Bigg(b \ \underset{qy}{\overset{c_1}{\underset{a_1'}{|}}} \ \underset{y}{\overset{c_2}{\underset{a_2'}{|}}} \ d \ x \Bigg)$$

$$= L_{\frac{y}{x}}(A, b, (c_k), d)$$

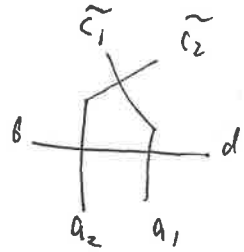
Proof of Prop 1 Apply YBE.

$$L_{\frac{y}{x}}(A, b, (c_k), d)\ (q+1)^{-\mathbb{1}_{a_1 \neq a_2}}$$

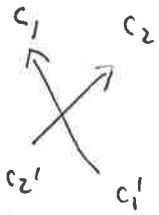
$=$ 

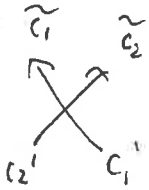
$=$ 

Take $\tilde{c}_k$ w/ $\tilde{c}_j = \{k : \tilde{c}_k = j\} = \{k : c_k = j\}$

$$L_{\frac{y}{x}}(A, b, (\tilde{c}_k), d)\ (q+1)^{-\mathbb{1}_{a_1 \neq a_2}} =$$ 

Projection property.

$$q^{-\mathrm{inv}(c_1, c_2)}$$ 

$$= q^{-\mathrm{inv}(\tilde{c}_1, \tilde{c}_2)}$$ 

$$\Rightarrow q^{-\mathrm{inv}(c_1, c_2)}\ (q+1)^{-\mathbb{1}_{a_1 \neq a_2}}\ L(A, b, (c_k), d)$$

$$= q^{-\mathrm{inv}(\tilde{c}_1, \tilde{c}_2)}\ (q+1)^{-\mathbb{1}_{a_1 \neq a_2}}\ L(A, b, (\tilde{c}_k); d)$$

General M. Suffices to show for a single flip.

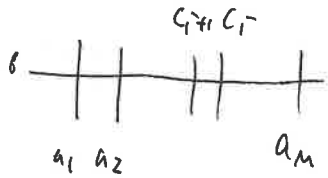
$$L_{\frac{y}{x}}(A, b, (c_1, c_2, \dots, c_M), d) = q^{-\mathrm{inv}(c_{i+1}, c_i)}\ L_{\frac{y}{x}}(A, b, (c_1, c_2, \dots c_{i-1}, c_{i+1}, c_i, \dots c_M), d)$$

$(*)$ $(**)$

$$(*) = \sum_{a_-, a_+} \sum_{a_i, a_{i+1}} \Big(b \ \underbrace{\overset{c_1}{|}\ \overset{c_2}{|}}_{a_-}\ \ \overset{a_i}{|}\ \overset{a_{i+1}}{|}\ \ \underbrace{\overset{c_M}{|}}_{a_+}\ d \Big)\ q^{\mathrm{inv}(a_i, a_{i+1})}\ q^{\mathrm{inv}(a_-, \{a_i, a_{i+1}\}) + \mathrm{inv}(a_+, \{a_i, a_{i+1}\})}$$

$\underline{\underline{M=2}}$ case

$$\sum_{a_-,\, a_+,\, \{a_i, a_{i+1}\}} \left(\sum_{a_i, a_{i+1}} \right.$$



$$\left. \right) q^{\mathrm{inv}(a_i, a_{i+1}) - \mathrm{inv}(c_i, c_{i+1})} \quad q^{\cdots}$$

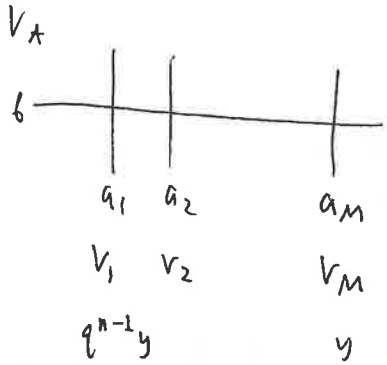
$= (**)$

<u>Remarks</u> 1) started w/ input $(a_1, a_2, \dots, a_M), b$

$$\underbrace{(e_{a_1} \otimes e_{a_2} \otimes \cdots \otimes e_{a_M})}_{\text{input vertical}} \otimes \underbrace{e_b}_{\text{input horizontal}}$$

$$\xrightarrow{q\text{-Symmetrize } (a_k)} \sum_{(a_k)} q^{\mathrm{inv}(a_k)} (e_{a_1} \otimes e_{a_2} \otimes \cdots \otimes e_{a_M}) \otimes e_b$$

$\xrightarrow{\text{Apply}}$



Apply $R^{M*}\left(\frac{y}{x}\right) \cdots R^{2*}\left(\frac{q^{M-2}y}{x}\right) R^{1*}\left(\frac{q^{M-1}y}{x}\right) \to \sum_{(a_k),(c_k)} q^{\mathrm{inv}(a_k)} R((a_k), b, (c_k), d)$

$e_{c_1} \otimes e_{c_2} \otimes \cdots \otimes e_{c_M} \otimes e_d$

<u>Claim</u> Output is q-symmetric $\Leftrightarrow$ Prop 1

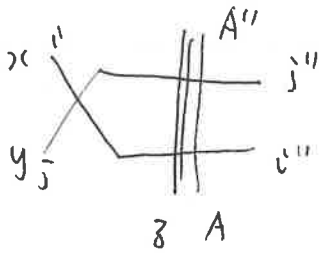
$\downarrow$ product of Rs acts on subspace of q-symmetric tensors

Generated by $\sum q^{\mathrm{inv}(c_k)} (e_{c_1} \otimes e_{c_2} \otimes \cdots \otimes e_{c_M}) \otimes e_d$

$z = y$

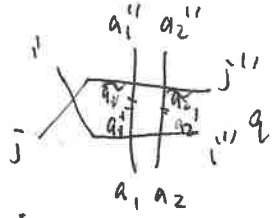
$$R^{2*}(z)\, R^{1*}(qz)\, R^{2*}(q) = R^{1*}(q)\, R^{2*}(z)\, R^{1*}(qz)$$

<u>Proof of Prop 2</u> (YBE)



$(\mu=2)$ ①

$$= \sum_{(a_1,a_2)} \; q^{\mathrm{inv}(a_1,a_2) - \mathrm{inv}(\cancel{a_1',a_2'})}$$



w/ constant A

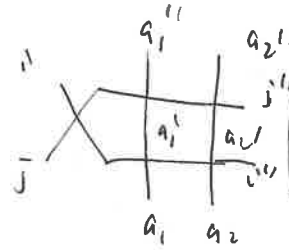
$(\tilde{a}_1, \tilde{a}_2)$

$q^{\mathrm{inv}(\cancel{\tilde{a}_1,\tilde{a}_2}) - \mathrm{inv}(a_1'', a_2'')}$

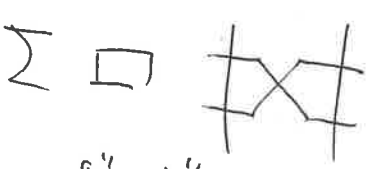
(a_1', a_2') anything.

Set equal to $(\tilde{a}_1, \tilde{a}_2)$

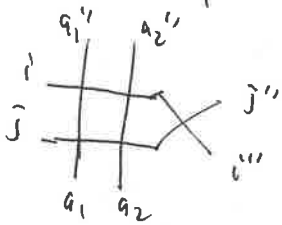
$$\overset{②}{=} \sum_{\substack{(a_1,a_2)\\(a_1',a_2')}} \; q^{\mathrm{inv}(a_1,a_2) - \mathrm{inv}(a_1'',a_2'')}$$



$$\overset{\text{YBE}}{=} \sum$$

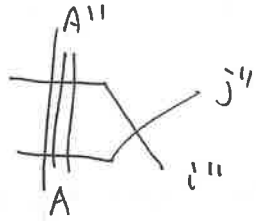


$$= \sum$$



$q^{\mathrm{inv}(a_1,a_2) - \mathrm{inv}(a_1'',a_2'')}$

$=$



Redoing ①, ②

Stochasticity

For $I = (I_0, I_1, \dots, I_n)$ w/ $\sum_{j=0}^{n} I_j = M$.

$$Z_q(I) = \frac{(q;q)_M}{(q;q)_{I_0}(q;q)_{I_1}\cdots(q;q)_{I_n}}$$ q-deformed multinomial coeff.

Lemma $$\sum_{\substack{(i_k) \text{ has} \\ \text{content } I}} q^{\operatorname{inv}(i_1, \dots, i_M)} = Z_q(I)$$ Proof by induction on M.

Define $$L^{\text{stoch}}_{\frac{y}{x}}(A, b, C, d) = \frac{Z_q(C)}{Z_q(A)} L_{\frac{y}{x}}(A, b, C, d)$$

Lemma $L^{\text{stoch}}_{\frac{y}{x}}(A, b, C, d)$ is stochastic

1) $$\sum_{C, d} L^{\text{stoch}}_{\frac{y}{x}}(A, b, C, d) = 1$$

2) L^{stoch} satisfy YBE.

Proof of 2) is same as proof of Prop 2. ← factors of $\frac{Z(C)}{Z(A)}$ cancel.

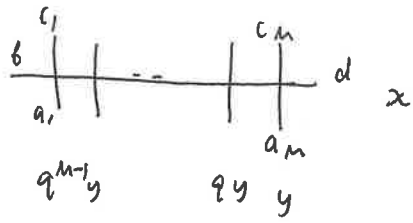
Proof of 1) (Outline)

R Stochastic $\sum_{c,d} R_z(a, b, c, d) = 1$.

→ row vertex is stochastic $\sum_{(c_k), d} R_z((a_k), b, (c_k), d) = 1$

→ put that into def'n of L and use Lemma

Lecture 6

$R_{\frac{y}{x}}\left((a_k), b, (c_k), d\right) =$ 

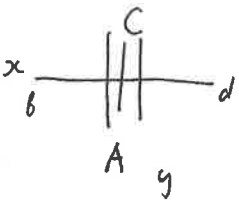
$$L^{\text{Stoch}}_{\frac{y}{x}}\left(A, b, (c_k), d\right) = \frac{Z_q(C)}{Z_q(A)} \sum_{\substack{(a_k) \sim \\ \text{content } A}} q^{\text{inv}(a_k)} R_{\frac{y}{x}}\left((a_k), b, (c_k), d\right)$$

$A \vdash \mathbb{Z}^{n+1}_{\geq 0} = (A_0, A_1, \cdots, A_n)$

$$Z_q(I) = \frac{(q;q)_{|I|}}{(q;q)_{I_0}(q;q)_{I_1}\cdots(q,q)_{I_n}} \qquad |I| = \sum_{j=0}^{n} I_j$$

Prop 1. $L^{\text{Stoch}}_{\frac{y}{x}}\left(A, b, (c_k), d\right) q^{-\text{inv}(c_k)}$ only depends on (c_k) through C.

$(*)$

$(*) = L^{\text{Stoch}}_{\frac{y}{x}}(A, b; C; d) =$ 

Prop 2. Satisfy YBE

Property: $\displaystyle\sum_{C,d} L^{\text{Stoch}}_{\frac{y}{x}}(A, b, C, d) = 1$

Today: Compute L^{Stoch}.

Theorem. Fix $A = (A_0, A_1, \dots, A_n)$, $C = (C_0, C_1, \dots, C_n)$, $\frac{y}{x} = z$

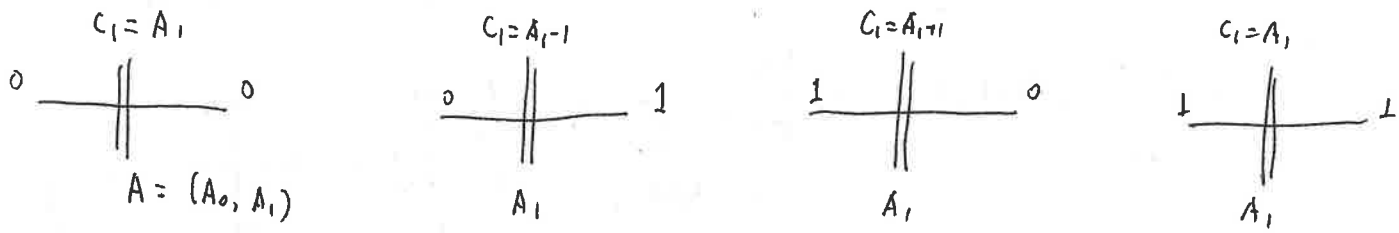
$$L_z^{stoch}(A, b, C, d) = \frac{1}{1 - q^M z} \begin{cases} (1 - q^{A_d} z)\, q^{A_{[d+1,n]}} & , b = d \\ (1 - q^{A_d})\, q^{A_{[d+1,n]}} & , b < d \\ (1 - q^{A_d})\, z\, q^{A_{[d+1,n]}} & , b > d \end{cases}$$

if array conservation holds. $\left(C = A^{+-}_{bd} \right)$

0 otherwise.

$$X_{[ij]} = \sum_{k=i}^{j} X_k$$

Example $n = 1 \rightsquigarrow b, d \in \{0, 1\}$



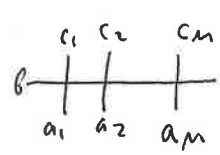
$A_0 = M - A_1$

$$\frac{(1 - q^{A_0} z)\, q^{A_1}}{1 - q^M z} = \frac{q^{A_1} - q^M z}{1 - q^M z} \qquad \frac{1 - q^{A_1}}{1 - q^M z} \qquad \frac{q^{A_1} z\,(1 - q^{M - A_1})}{1 - q^M z} = \frac{q^{A_1} z - q^M z}{1 - q^M z} \qquad \frac{1 - q^{A_1} z}{1 - q^M z}$$

Set $s^2 = q^{-M}$, $z \rightsquigarrow z^{-1}$ recovers L^{stoch} first / second class

Proof of Theorem.

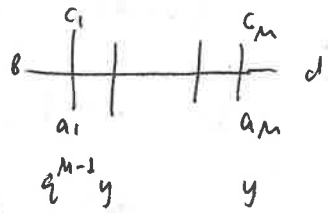
Case $b = d$. $L_z^{stoch}(A, b, C, d) = \overset{1}{\overset{\shortparallel}{\boxed{\frac{Z_q(C)}{Z_q(A)}}}} \sum_{\substack{a_k \\ \text{count } A}}$ pick (C_k) w/ count C $\; q^{inv(a_k)}$



$q^{-inv(c_k)}$

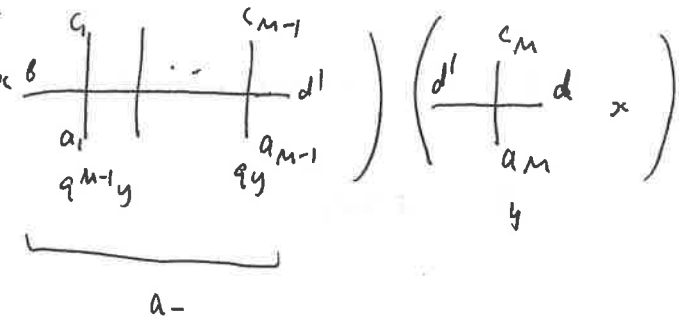
Pick c ordered so that $c_1 \le c_2 \le \cdots \le c_M$ $\rightsquigarrow$ $q^{\mathrm{inv}(c_k)} = 1$

$$L_{\mathfrak{z}}^{\mathrm{stoch}}(A, b, c, d) = \sum_{(a_k)} q^{\mathrm{inv}(a_k)}$$



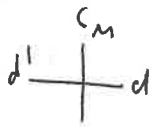
want to turn this into L

Recurse on M $=$

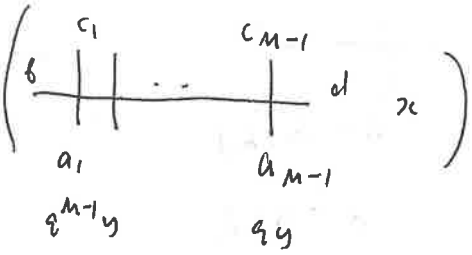


$$\mathrm{inv}(a_k) = \mathrm{inv}(a_-) + \mathrm{inv}(a_-, a_M)$$

$$= \sum_{a_M} q^{\mathrm{inv}(a_-, a_M)}$$



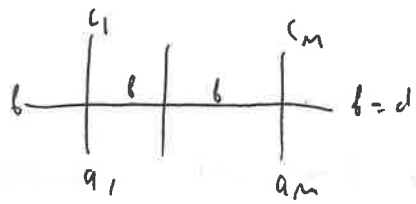
$$\sum_{a_1, \ldots, a_{M-1}} q^{\mathrm{inv}(a_-)}$$



$$\| \quad \frac{Z_q(A^-)}{Z_q(C^-)} L_{q\mathfrak{z}}^{\mathrm{stoch}}(A^-, b, C^-, d')$$

Expresses $L_{\mathfrak{z}}^{\mathrm{stoch}}(A, b, C, d)$ w/ M in terms of $L_{q\mathfrak{z}}^{\mathrm{stoch}}(\tilde{A}, \tilde{b}, \tilde{c}, \tilde{d})$ w/ $M-1$.

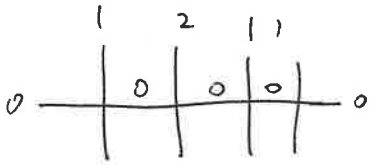
<u>Thought</u>: if $(c_k) = (a_k)$, then easier



<u>Issue</u>: only fix (c_k) after sum is taken

$\rightsquigarrow$ so it <u>cannot</u> depend on (a_k).

Suppose that (c_k) had no $d = b$ $\longrightarrow$ All horizontal edges would have to be $b = d$.



Some perm. of $\{1,1,1,2\}$

$\longrightarrow$ Sum would only have one term for (a_k)

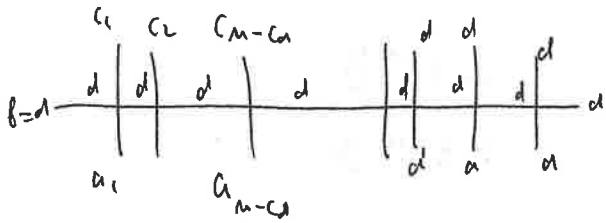
We would have to have $(a_k) = (c_k)$

Address possible bs in c

Choose bs to be to the left (almost work)

<u>right</u> (produces one term)

<u>Simplification</u> Choose (c_k) s.t. $c_M = c_{M-1} = \dots = c_{M-c_d+1} = d$ (push all ds to the right)



$a_1 = c_1$

$a_2 = c_2$

$\vdots$

$a_{M-c_d} = c_{M-c_d}$

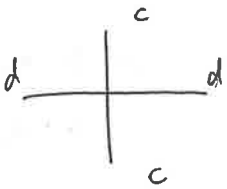
$\vdots$

$a_M = c_M$

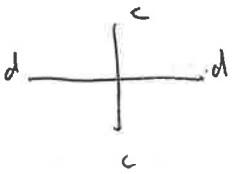
$\longrightarrow$ Sum over (a_k) is <u>frozen</u> on $(a_k) = (c_k)$

$$L^{\text{Stoch}}_{\frac{y}{x}}(A, b, c, d) = \overbrace{\boxed{\frac{Z_q(c)}{Z_q(A)}}}^{1} q^{-\mathrm{inv}(c_k)} \sum_{(a_k)} q^{\mathrm{inv}(a_k)} \Big(b \xrightarrow{c_1 \; c_2 \; \cdots \; c_M} b = d \Big)$$

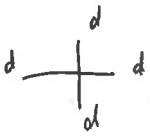
$$= q^{-\mathrm{inv}(c_k)} \, q^{\mathrm{inv}(c_k)} \; d \xrightarrow[q^{M-1}y \;\; qy \;\; y]{c_1 \; c_2 \; \cdots \; c_M} d \quad x$$

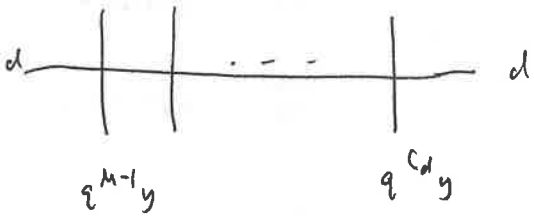


$c > d$, $\quad \dfrac{(1-z)q}{1-qz}$



$c < d$ $\quad \dfrac{1-z}{1-qz}$

Rightmost C_d vertices are 

$=$ 

$$= q^{\#\{c_j > d\}} \frac{1 - q^{M-1}\frac{y}{x}}{1 - q^{M}\frac{y}{x}} \cdot \frac{1 - q^{M-2}\frac{y}{x}}{1 - q^{M-1}\frac{y}{x}} \cdots \frac{1 - q^{C_d}\frac{y}{x}}{1 - q^{C_d+1}\frac{y}{x}}$$

$$= q^{\#\{c_j > d\}} \frac{1 - q^{C_d} z}{1 - q^{M} z} \quad \checkmark$$

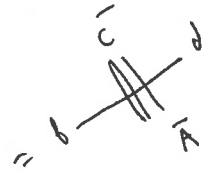
The other two cases are similar. (push b or d to the left or right in (C_k))

More than one configuration for (a_k), but the sum will telescope.

<u>Analytic continuation of q^M</u>

$A = (A_0, A_1, \dots, A_n)$ satisfied $\sum_{i=0}^{n} A_i = M \rightsquigarrow A_0 = M - \sum_{i=1}^{n} A_i$

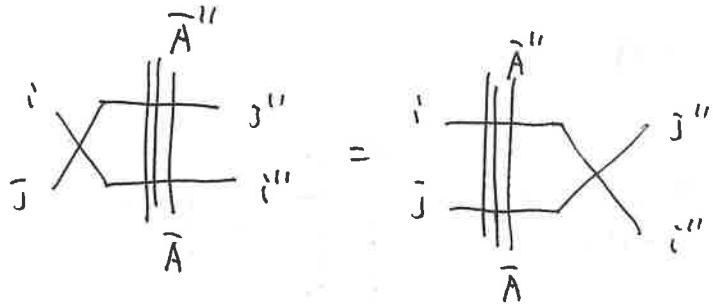
(because M columns) Track $\bar{A} = (A_1, \dots, A_n)$ instead.



For any $M \in \mathbb{C}$, $\bar{A}, \bar{c} \in \mathbb{Z}_{\geq 0}^{n}$, define $L_z^{Stoch}(A, b, c, d) = (*)$ where $A_0 = M - \sum_{i=1}^{n} A_i$

$= M - A[1, n]$

Prop 3



$(**)$

Proof. Original YBE implies $(**)$ if $M \in \mathbb{Z}_{\geq 1}$.

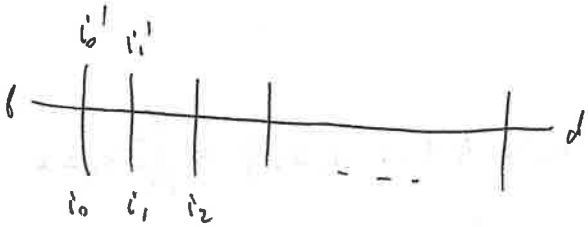
Also, $(**)$ is equality of rat'l functions in q^M $\longrightarrow$ true for all $q^M \in \mathbb{C}$

$q^M = s^{-2} \in \mathbb{C}$

Σ

Row operators

For most of what follows, we take $\underline{n=1}$.



Associate w/ this row an operator that "acts" on inputs $(i_0, i_1, i_2, \dots)$ and outputs a linear combination of "outputs" $(i_0', i_1', \dots)$

Define vector spaces

$(V_i)_{i \geq 0}$

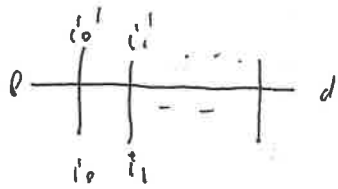
$V_i = \text{Span}\{\overset{e_k}{|k\rangle}\}_{k \geq 0}$

Think of k as indexing # of arrows entering at i

$V = V_0 \otimes V_1 \otimes \cdots$ = spanned by $|i_0\rangle \otimes |i_1\rangle \otimes \cdots \otimes |i_n\rangle \otimes \cdots$ $= |\vec{i}\rangle$

We'll define <u>row operators</u> $T_{bd} : \mathbb{V} \longrightarrow \mathbb{V}$

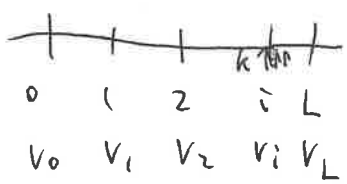
$$T_{bd} |\vec{i}\rangle = \sum_{\vec{i}'} |\vec{i}'\rangle$$



Many properties of symmetric functions can be read off of algebraic relations between row operators.

<u>Lecture 7</u>. For each integer $i \geq 0$, let $V_i = \mathrm{Span}\{|k\rangle\}_{k \geq 0}$

Think of V_i as indexing the ith column of a vertex model



$k = \#$ arrows coming in

For any $\lambda = (\lambda_1, \lambda_2, \cdots, \lambda_\ell) = 1^{m_1} 2^{m_2} \cdots$ $\in \mathbb{Z}_{\geq 0}^{\ell}$, $\quad m_i = \#\{j : \lambda_j = i\} \Rightarrow \sum_{i=1}^{\infty} m_i = \ell$

Define $|\lambda\rangle = \underset{\in V_0}{|m_0\rangle} \otimes \underset{\in V_1}{|m_1\rangle} \otimes \cdots \in V_0 \otimes V_1 \otimes \cdots$

For any $L \geq 1$, if $\lambda_1 \leq L$, $|\lambda\rangle_L = |m_0\rangle \otimes |m_1\rangle \otimes \cdots \otimes |m_L\rangle \in V_0 \otimes V_1 \otimes \cdots \otimes V_L$

Observe that for $L' > L$, $|\lambda\rangle_{L'} = |\lambda\rangle_L \otimes |m_{L+1}\rangle \otimes |m_{L+2}\rangle \otimes \cdots \otimes |m_{L'}\rangle$

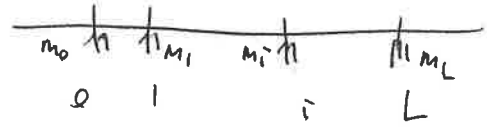
$$= |\lambda\rangle_L \otimes |0\rangle \otimes |0\rangle \otimes \cdots \otimes |0\rangle$$

so morally the same

Define $\langle \mu | \lambda \rangle = \mathbb{1}_{\mu = \lambda}$, $\langle \mu | \lambda \rangle_L = \mathbb{1}\{m_i(\mu) = m_i(\lambda), \forall i \in [0, L]\}$

so have orthonormal basis indexed by partitions.

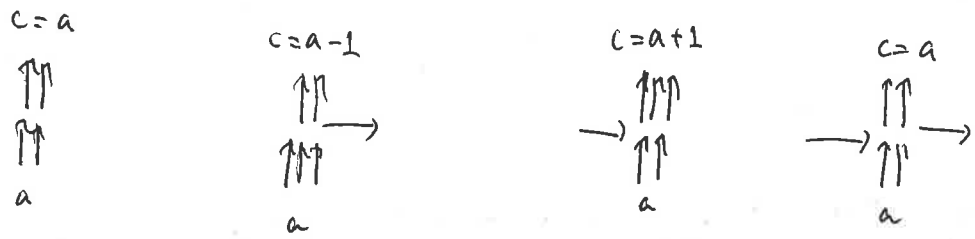
Think of $|\lambda\rangle$ as indexing an infinite row



$w(a,b,c,d)$ Spin Hall-Littlewood weights. are equal to 0 unless $b, d \in \{0,1\}$

row spectral param. $x \longrightarrow$

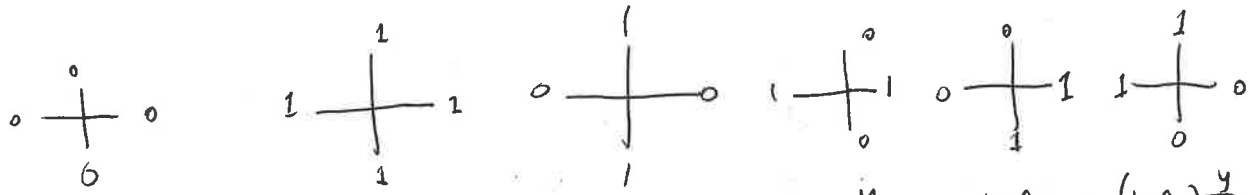
(column $y = 1$)



$$\frac{1-sq^a x}{1-sx} \qquad \frac{(1-s^2q^{a-1})x}{1-sx} \qquad \frac{1-q^{a+1}}{1-sx} \qquad \frac{x-sq^a}{1-sx}$$

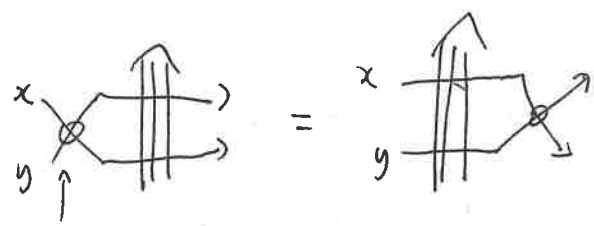
Satisfy YBE

6 vertex weights



$$1 \qquad 1 \qquad \frac{q(1-\frac{y}{x})}{1-q\frac{y}{x}} \qquad \frac{1-\frac{y}{x}}{1-q\frac{y}{x}} \qquad \frac{1-q}{1-q\frac{y}{x}} \qquad \frac{(1-q)\frac{y}{x}}{1-q\frac{y}{x}}$$

YBE



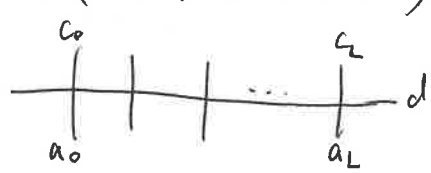
6 vertex wts

horizontal rapidity x, vertical rapidity y

<u>Row operators</u> (first for truncated, then for infinite)

Define the row weight

$$W_u(\{a_k\}, b, \{c_k\}, d) \qquad \{a_k\}, \{c_k\} \in \mathbb{Z}^{L+1}_{\geq 0}$$

$= b$  d

$u \to$ rapidity

(like fusion but $a_k > 1$ possible)

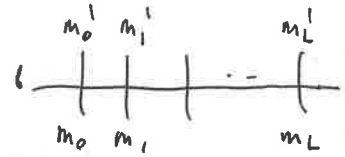
with spin HL weights

unique configuration given a, b, c, d w/ arrow conservation

Define $T_u(b,d) = T_u^L(b,d) : V_0 \otimes \cdots \otimes V_L \to V_0 \otimes \cdots \otimes V_L$

$$T_u(b,d)\underset{\substack{\uparrow \\ |m_0\rangle \otimes \cdots \otimes |m_L\rangle}}{|\lambda\rangle_L} = \sum_{\{m_k'\}} W_u\left(\{m_k\}, b, \{m_k'\}, d\right) \underset{\substack{\uparrow \\ |m_0'\rangle \otimes \cdots \otimes |m_L'\rangle}}{|\lambda'\rangle}$$

$$= \sum_{m'}$$



$|\lambda'\rangle$

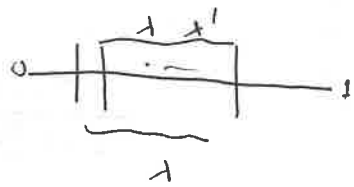
monodromy matrix (from quantum integrable systems)

$$T(u) = T^L(u) = \begin{bmatrix} T_u(0,0) & T_u(0,1) \\ T_u(1,0) & T_u(1,1) \end{bmatrix} = \begin{bmatrix} A_L(u) & B_L(u) \\ C_L(u) & D_L(u) \end{bmatrix}$$

$A_L(u)|\lambda\rangle \to \sum_{\lambda'}$

$|\lambda'\rangle$

$B_L(u)|\lambda\rangle \to \sum_{\lambda'}$



$|\lambda'\rangle$

- - -

These (A_L, B_L, C_L, D_L) satisfy commutation relations due to YBE.

lots of them but 3 main ones we will use: (should have for all pairs)

<u>Prop</u> $u, v \in \mathbb{C}$ (rapidities)

1) $A_L(u)A_L(v) = A_L(v)A_L(u)$, same for B, C, D : self-commuting

($\rightsquigarrow$ gives rise to symmetry of functions eventually)

2) $\left(1 - \frac{qu}{v}\right) C_L(u) D_L(v) = \left(1 - \frac{u}{v}\right) D_L(v) C_L(u) + (1-q) C_L(v) D_L(u)$

(some form of $[C, D]$)

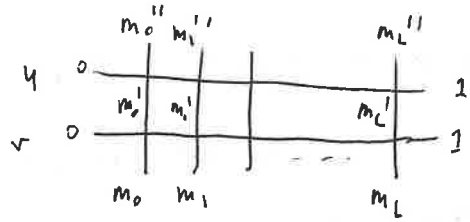
3) $\left(1 - \frac{qu}{v}\right) C_L(v) A_L(u) = \left(1 - \frac{u}{v}\right) A_L(u) C_L(v) + (1-q)\frac{u}{v} C_L(u) A_L(v)$

Proof

1) For B WTS $B_L(u)\, B_L(v)\, |\lambda\rangle = B_L(v)\, B_L(u)\, |\lambda\rangle$

Draw through vertex model

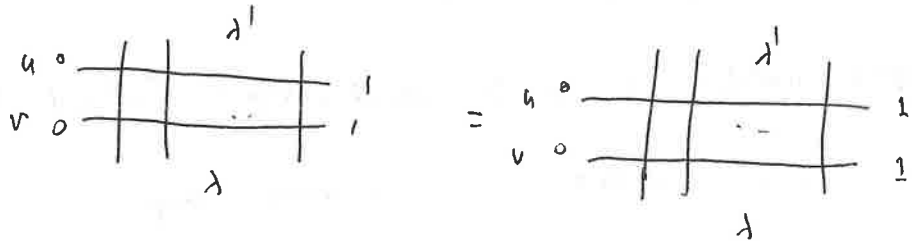
$\sum_{\lambda', \lambda''}$



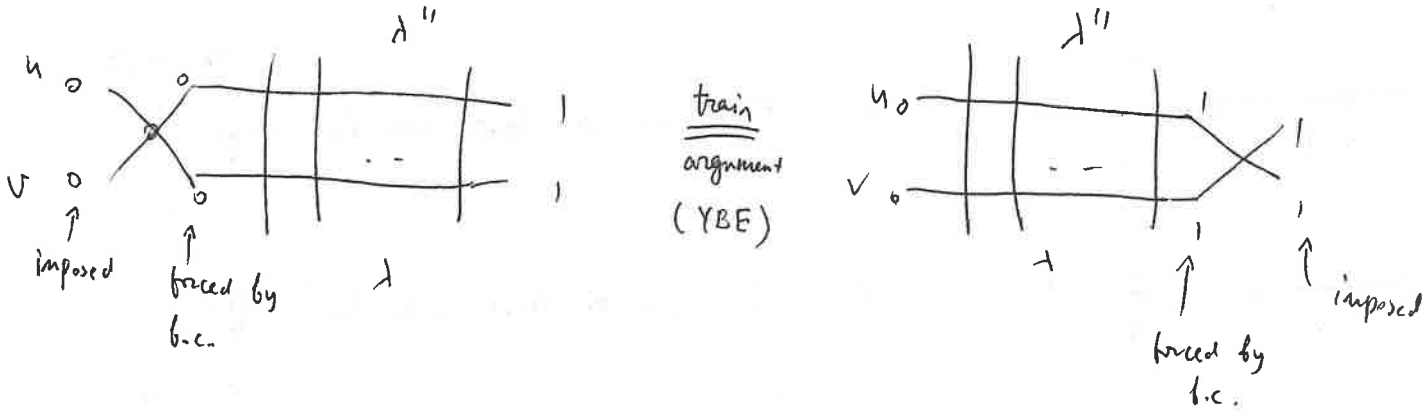
$|\lambda''\rangle$

Want $\langle \lambda'' | B_L(u)\, B_L(v) | \lambda \rangle = \langle \lambda'' | B_L(v)\, B_L(u) | \lambda \rangle$

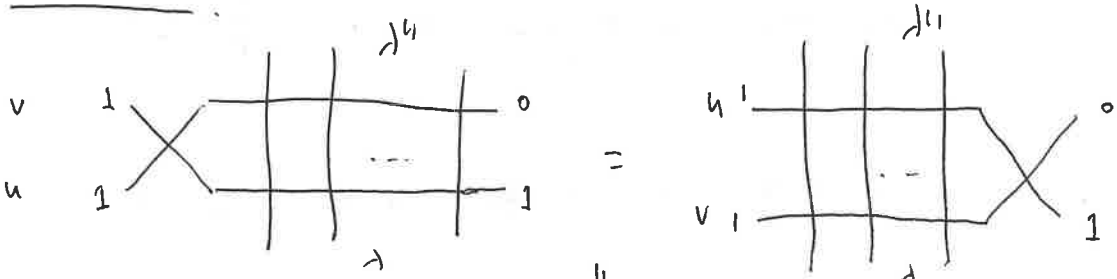
i.e.



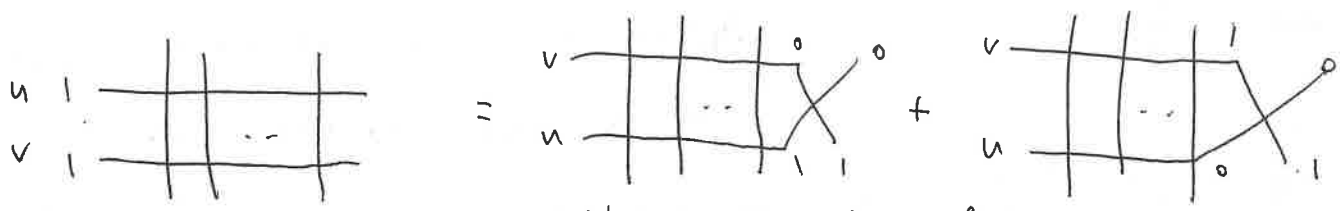
Attach a 6V cross ⊗ to left



Proof of (2)



$\Downarrow$



$$\langle \lambda'' | C_L(u)\, D_L(v) | \lambda \rangle = \langle \lambda'' | C_L(v)\, D_L(u) | \lambda \rangle \frac{1-q}{1-q\frac{u}{v}} + \langle \lambda'' | D_L(v)\, C_L(u) | \lambda \rangle \frac{1-\frac{u}{v}}{1-q\frac{u}{v}}$$

Clear denominators

$$\rightarrow \left(1-\frac{qu}{v}\right) C_L(u) D_L(v) = \left(1-\frac{u}{v}\right) D_L(v) C_L(u) + (1-q) C_L(v) D_L(u)$$

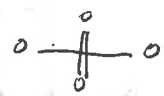
$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ is quantum monodromy matrix: limit of some classical object tracking monodromy of solutions in integrable Hamilt. dynamics.

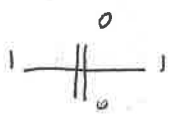
YBE: $R\left(\frac{u}{v}\right) T(u) \otimes T(v) = T(v) \otimes T(u) R\left(\frac{u}{v}\right)$ (RTT = TTR relation)

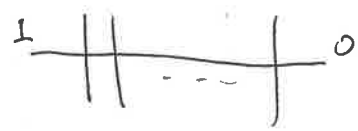
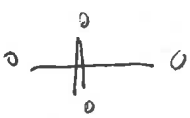
$\hookrightarrow$ equality of 4×4 matrices w/ operator valued entries $T_u(b,d)\, T_v(b',d')$

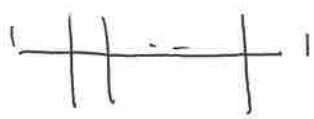
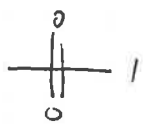
algebra of ... / ... relations $\longrightarrow$ quantum group

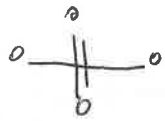
Infinite volume limit $(L \to \infty)$ want # columns $\to \infty$, input / output are finite.

A all vertices far to right will be 

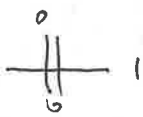
B all vertices far to right will be 

C  all vertices far to right will be 

D  all vertices far to right will be 

Recall weight  $= 1$ $\longrightarrow$ $A_L(u)$, $C_L(u)$ are well-defined as $L \to \infty$

$A_\infty(u) = A(u)$, $C_\infty(u) = C(u)$

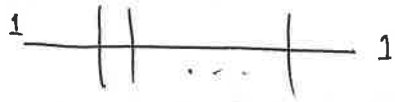
u  $= \frac{u-s}{1-su}$ $\longrightarrow$ $B_L(u)$, $D_L(u)$ contain factors of $\left(\frac{u-s}{1-su}\right)^{L+1}$

Define $\hat{B}_L(u) = \left(\frac{1-su}{u-s}\right)^{L+1} B_L(u)$, $\hat{D}_L(u) = \left(\frac{1-su}{u-s}\right)^{L+1} D_L(u)$

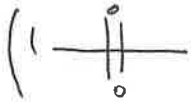
$\hat{B}_L$, $\hat{D}_L$ can be interpreted as the weights of a row

$\hat{B}$

$\hat{D}$



w/ slightly different weights $\hat{w}(a,b,c,d) = w(a,b,c,d)\left(\frac{1-su}{u-s}\right)$

$\rightsquigarrow \hat{w}\Big($  $\Big) = 1$

$\hat{B}_L(u)$, $\hat{D}_L(u)$ well-defined as $L \to \infty$ $\rightsquigarrow$ $B(u) = \hat{B}_\infty(u)$, $D(u) = \hat{D}_\infty(u)$

$L \to \infty$ simplifies some commutation relations.

<u>Prop.</u> Assume $u, v, s \in \mathbb{C}$ satisfy $\left|\frac{(u-s)(v-s)}{(1-su)(1-vs)}\right| < 1$, then

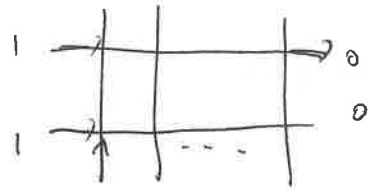
$$(1-quv)\, C(u)\, D(v^{-1}) = (1-uv)\, D(v^{-1})\, C(u).$$

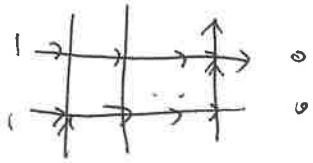
$$(1-quv)\, C(u)\, D(v^{-1}) = (1-quv) \lim_{L\to\infty} C_L(u)\, D_L(v^{-1}) \left(\frac{1-\frac{s}{v}}{\frac{1}{v}-s}\right)^{L+1}$$

$$\overset{(2)}{\underset{w/\, v\to v^{-1}}{=\!=}} \lim_{L\to\infty} \left((1-uv)\, D_L(v^{-1})\, C_L(u) \left(\frac{v-s}{1-sv}\right)^{L+1} + (1-q)\, C_L(v^{-1})\, D_L(u) \left(\frac{v-s}{1-sv}\right)^{L+1} \right)$$

$$= (1-uv)\, D(v^{-1})\, C(u) + \lim_{L\to\infty} \underbrace{(1-q)\, C_L(v^{-1})\, D_L(u) \left(\frac{1-su}{u-s}\right)^{L+1}}_{\downarrow\ (1-q)\, C(v^{-1})\, D(u)} \underbrace{\left(\frac{(v-s)(u-s)}{(1-vs)(1-us)}\right)^{L+1}}_{\downarrow\ 0}$$

need $|\cdots| < 1$ for things to be well-defined.

Consider 

then will have  w/ nonvanishing weight

so $L \to \infty$ limit not well-defined.

technical issue – operators defined on $(V_0 \otimes V_1 \otimes \cdots)^{fin}$: only finitely many factors not $|0\rangle$

each A, B, C, D well-defined but not e.g. CD always because you need to sum over middle terms

Maybe one way to think of it: domain of A, B, C, D is $\bigcup_{L \geq 0} V_0 \otimes \cdots \otimes V_L$

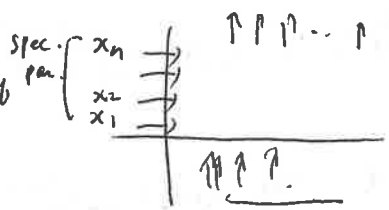
so a priori range of A, B, C, D is bigger and can't apply A, B, C, D again.

but w/ some decay condition $|\cdots| < 1$, the operators CD, etc. can still be defined.

<u>Lecture 8</u>. <u>Def'n</u> Take signatures ← elements of $\mathbb{Z}^{\ell}_{\geq 0}$ for some ℓ

λ, μ (might or might not be same length)

1) $F_{\lambda|\mu}(x_1, x_2, \cdots, x_n) = \langle \lambda | C(x_n)\, C(x_{n-1}) \cdots C(x_1) | \mu \rangle$

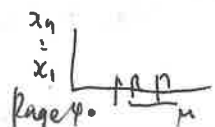
$= $ part. func. of 

↑↑↑.. ↑ indexed by λ

indexed by μ

Nonzero only if $\ell(\lambda) = \ell(\mu) + n$.

2) $h_{\lambda|\mu}(x_1, x_2, \cdots x_n) = \langle \lambda | A(x_n)\, A(x_{n-1}) \cdots A(x_1) | \mu \rangle$

$=$ part. func. of 

nonzero only if $\ell(\lambda) = \ell(\mu)$.

(Similar for B, D, other bdry conditions)

These will be symmetric functions that satisfy many of the properties in the first class:

1) Combinatorial formula (def'n)

2) Symmetric in (x_i)

3) Branching identity (how to recurse in n)

4) Cauchy identity

5) Symmetrization formula

6) Littlewood-Richardson

(determinantal only at $q=0$)

<u>Prop (2)</u> $F(x)$, $G(x)$ are sym. in x.

$$G_{\lambda|\mu}(x_1, x_2, \cdots x_n) = \langle \lambda | A(x_n) A(x_{n-1}) \cdots A(x_1) | \mu \rangle$$

Recall $A(u) A(v) = A(v) A(u)$ (*)

$$\overset{(*)}{=} \langle \lambda | A(x_{\sigma(n)}) A(x_{\sigma(n-1)}) \cdots A(x_{\sigma(1)}) | \mu \rangle$$

$$= G_{\lambda|\mu}(x_{\sigma(1)}, x_{\sigma(2)}, \cdots, x_{\sigma(n)})$$

(implicit train argument in commutation relations)

Proof for F the same, use $C(u)C(v) = C(v)C(u)$.

<u>Branching</u>

<u>Prop</u> $$F_{\lambda|\mu}(x_1, \cdots, x_n) = \sum_{\nu} F_{\lambda|\nu}(x_{m+1}, x_{m+2}, \cdots x_n) F_{\nu|\mu}(x_1, x_2, \cdots, x_m)$$

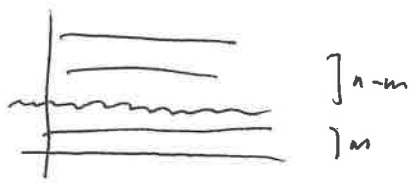
$$G_{\lambda|\mu}(x_1, \cdots, x_n) = \sum_{\nu} G_{\lambda|\nu}(x_{m+1}, x_{m+2}, \cdots, x_n) G_{\nu|\mu}(x_1, x_2, \cdots, x_m)$$

Proof for F (G same)

$$F_{\lambda|\mu}(x_1, x_2, \dots, x_n) = \langle \lambda | C(x_n) C(x_{n-1}) \cdots C(x_1) | \mu \rangle$$

$$= \langle \lambda | C(x_n) C(x_{n-1}) \cdots C(x_{m+1}) \; C(x_m) \cdots C(x_1) | \mu \rangle$$

insert $Id = \sum_\nu |\nu\rangle\langle\nu|$



$$= \sum_\nu \langle \lambda | C(x_n) \cdots C(x_{m+1}) | \nu \rangle \langle \nu | C(x_m) \cdots C(x_1) | \mu \rangle$$

$$= \sum_\nu F_{\lambda|\nu}(x_{m+1}, \dots x_n) \, F_{\nu|\mu}(x_1, \dots x_m).$$

Cauchy identity: Set $G^c_{\lambda|\mu}(x_1, x_2, \dots, x_n)$ (c = conjugation)

$$= \frac{c_\lambda}{c_\mu} G_{\lambda|\mu}(x_1, \dots, x_n)$$

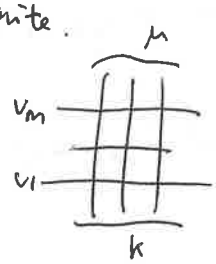
$$c_\nu = \prod_{k \geq 0} \frac{(s^2; q)_{m_k(\nu)}}{(q;q)_{m_k(\nu)}}$$

Thm Let $(u_1, u_2, \dots, u_\ell)$, $(v_1, v_2, \dots, v_m)$ be cplx numbers s.t.

$\left| \frac{(u_i - s)(v_j - s)}{(1 - s u_i)(1 - s v_j)} \right| < 1, \; \forall i, j.$ then for any $\mu, \nu \in \bigcup_{\ell \geq 0} \mathbb{Z}^\ell_{\geq 0}$,

$$\sum_\lambda F_{\lambda|\mu}(u) \, G^c_{\lambda|\nu}(v) = \sum_k F_{\nu|k}(u) \, G^c_{\mu|k}(v) \prod_{i,j} \frac{1 - q u_i v_j}{1 - u_i v_j}$$

Rmk. On left side, sum over λ genuinely infinite, but on the right side, it is really finite.

$G_{\mu|k} =$ 

$= 0$ unless $k \subset \mu \iff k_i \leq \mu_i, \forall i \geq 0$

$\Rightarrow$ only finitely many k for which $G_{\mu|k} \neq 0$.

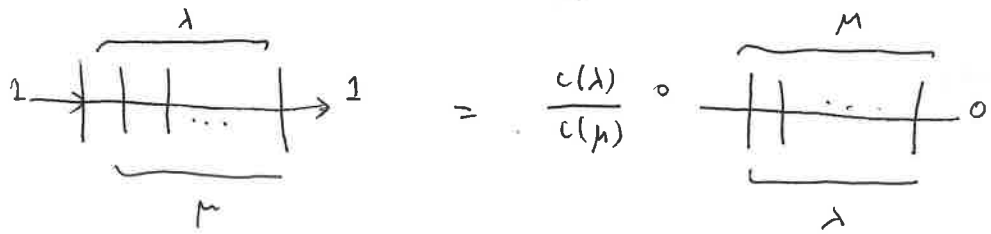
Proof Want to use $(1-quv)\,C(u)\,D(v^{-1}) = (1-uv)\,D(v^{-1})\,C(u)$

C defines F, A defines G

(could have used D for G, but different boundary conditions)

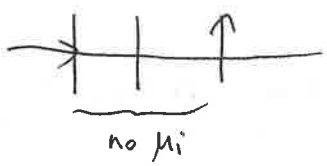
Fact $\langle\lambda|\,D(u)\,|\mu\rangle = \dfrac{c(\lambda)}{c(\mu)}\,\langle\mu|\,A(u^{-1})\,|\lambda\rangle$

Proof. Suffices to check the following:



first arrow exists somewhere,

μ must enter after.

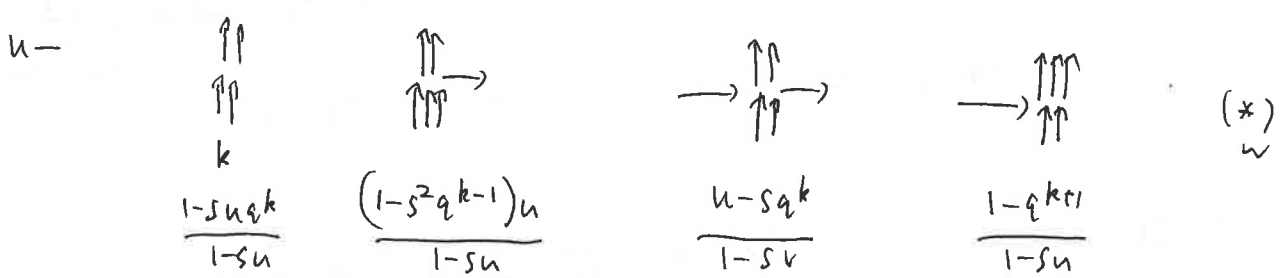


smallest $\lambda \le$ smallest μ

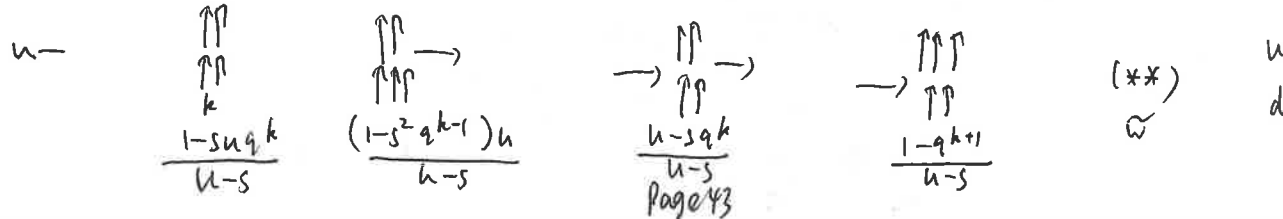
$\Rightarrow \lambda_i \le \mu_i$ (induct)

reversed: $\lambda_i \le \mu_i$.

Pf'n of A use vertex weights

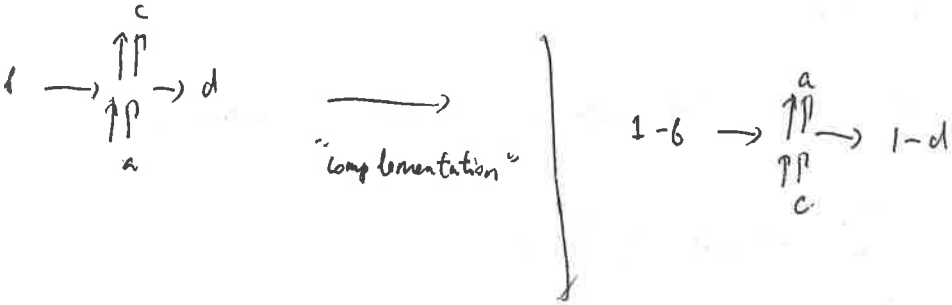


D_L was also defined using $(*)$. However to define D, we needed to multiply each weight by $\dfrac{1-su}{u-s}$ $\rightsquigarrow$ gives rise to



use $(**)$ to define D.

$$\tilde{W}_n(a,b;c,d) = \frac{(s^2;q)_c}{(q;q)_c}\,\frac{(q;q)_a}{(s^2;q)_a}\, W_{n-1}(c,1-b,a,1-d) \qquad (***)$$



picks up $\frac{c_\lambda}{c_\mu}$

Pf check case by case.

Use $(***)$ at each vertex.

(same for B, C)

<u>Proof of theorem</u>

$$\sum_\lambda F_{\lambda|\mu}(u)\, G^c_{\lambda|\mu}(v) = \sum_\lambda F_{\lambda|\mu}(u)\, G_{\lambda|\nu}(v)\,\frac{c(\lambda)}{c(\nu)}$$

$$= \sum_\lambda \langle\lambda|C(u_\ell)\cdots C(u_1)|\mu\rangle\langle\lambda|A(v_m)\cdots A(v_1)|\nu\rangle\,\frac{c(\lambda)}{c(\nu)}$$

$$\overset{\text{Fact}}{=} \sum_\lambda \langle\lambda|C(u_\ell)\cdots C(u_1)|\mu\rangle\langle\nu|D(v_1^{-1})\cdots D(v_m^{-1})|\lambda\rangle$$

(telescope by resolving identity repeatedly)

Use : $(1-quv)\,C(u)\,D(v^{-1}) = (1-uv)\,D(v^{-1})\,C(u)$

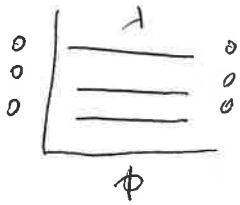
$$= \prod \frac{1-qu_iv_j}{1-u_iv_j}\,\langle\lambda|D(v_1^{-1})\cdots D(v_m^{-1})|\mu\rangle\langle\nu|C(u_\ell)\cdots C(u_1)|\lambda\rangle$$

$$= \prod \frac{1-qu_iv_j}{1-u_iv_j}\sum_\lambda G^c_{\mu|\lambda}(v)\,F_{\nu|\lambda}(u)$$

Take $\mu=\phi$, $\nu=0^m$. $\quad \sum \underbrace{G^c_{\lambda|0^m}(v)}_{G_{\lambda|0^m}=G_\lambda \text{ pageway}}\,\underbrace{F_{\lambda|\phi}(u)}_{u}$ λ

$F_{\lambda|\phi}=F_\lambda$

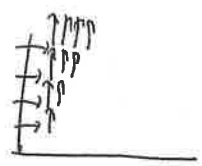
$$= \prod \frac{1-qu_iv_j}{1-u_iv_j} \sum_\lambda \underbrace{G^c_{\phi|\lambda}(v)}_{} \; F_{0m|\lambda}(u)$$



$\Rightarrow \lambda = \phi$

Supported on $\lambda = \phi$, $G_{\phi|\phi} = 1$

$$= \prod \frac{1-qu_iv_j}{1-u_iv_j} \underbrace{F_{0m|\phi}(u)}_{}$$



factors

So gives completely factored Cauchy identity

<u>Lecture 9</u>. <u>Last time</u>: Fix μ, ν signature $\leftarrow$ (sequences of nonnegative integers)

Let $(u_i), (v_j) \in \mathbb{C}$ s.t. $\left| \frac{(u_i - s)(v_j - s)}{(1-su_i)(1-sv_j)} \right| < 1, \quad \forall i,j$

$$\rightsquigarrow \sum_\lambda F_{\lambda|\mu}(u_1, \dots, u_m) \; G^c_{\lambda|\nu}(v_1, \dots, v_n)$$

<u>Skew</u> - Cauchy identity.

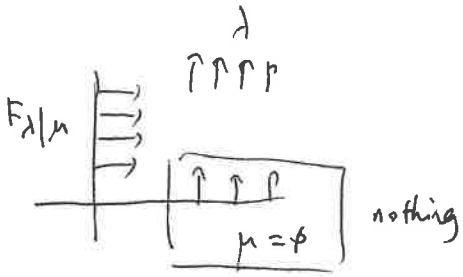
$$= \prod_{i,j} \frac{1-qu_iv_j}{1-u_iv_j} \sum_k F_{\nu|k}(u_1, \dots, u_m) \; G^c_{\mu|k}(v_1, \dots, v_n) \qquad (*)$$

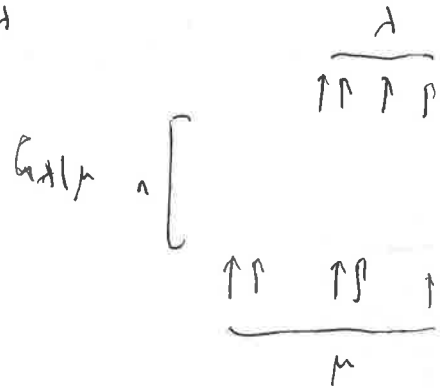
<u>Rmk</u> LHS infinite sum, RHS finite

<u>Today</u>. Simplification of skew-Cauchy $\longrightarrow$ Cauchy

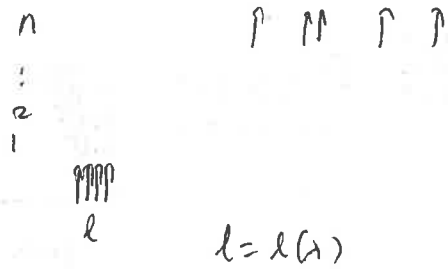
Towards probabilistic applications

<u>Notation</u> When $\mu = \phi$, $F_{\lambda|\phi} =: F_\lambda$





If $\mu = 0^\ell \rightsquigarrow G_{\lambda|0^\ell} =: G_\lambda$

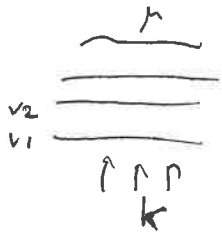


$\ell = \ell(\lambda)$

<u>Cauchy</u>. Factored form for $\sum_\lambda F_\lambda(u)\, G_\lambda(v)$ $\quad (**)$

To turn LHS of $(*)$ into $(**)$, set $\mu = \phi$, $\nu = 0^\ell$.

RHS has $G_{\mu|k}$



If $\mu = \phi$, nothing leaves

$\rightarrow k = \phi$

weight of $\circ\!\!\!

product from $k=0$ to $k=m-1$

$$F_{0m}(u) = \prod_{k=1}^{m} \frac{1-q^k}{1-su_k} = (q;q)_m \prod_{k=1}^{m} (1-su_k)^{-1}$$

$$\sum_{\lambda} F_\lambda(u_1, \dots, u_m)\, G_\lambda^c(v_1, \dots, v_n) = \prod_{i,j} \frac{1-qu_iv_j}{1-u_iv_j} (q;q)_m \prod_{i=1}^{m} (1-su_i)^{-1}$$

Rmk. set $s=0$, $q=0$ $\rightsquigarrow$ F_λ, G_λ $\rightsquigarrow$ Schur poly.

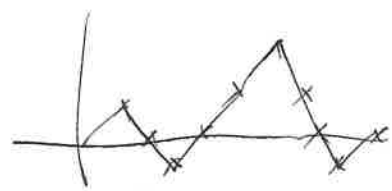
set $s=0$, $q\neq 0$ $\quad$ F_λ, G_λ $\rightsquigarrow$ HL poly.

Random walk

$X_0 = 0$

$X_i = X_{i-1} + B_i$

$B_i \in \{-1, 1\}$ Random



$X(i) = X_i$ is a random function on $\mathbb{Z}_{\geq 0}$ s.t. $|X(i) - X(j)| = 1$ if i is adjacent to j.

(Random) path

Naturally sample random path in two ways

1) "Markovian" way

2) "Bridged" way.

1) Know $X(i)$, sample $X(i+1) - X(i)$ independent of $(X(0), X(1), \dots, X(i-1))$

"generating walk"

2) Fix $X(0) = 0$, $X(N) = K$,] look at all paths $|X(i) - X(j)| = 1$ if i,j adjacent.

"Random walk bridge" $\quad$ w/ $X(0) = 0$, $X(N) = K$

Central limit theorem

In cases (1), (2), $N^{-1/2} X(N) \longrightarrow$ Gaussian random variable

(1) $\longrightarrow N^{-1/2} X(tN) \longrightarrow G(t)$ Brownian motion

(2) $\longrightarrow N^{-1/2} X(tN) \longrightarrow B(t)$ Brownian bridge

Behaviour of a random ~~path~~ curve

1) Stochastic generation of curve

2) "Equilibrium generation" of curve

How to do this in higher dimensions

Want random surface

Want analogs of CLTs

~~Originally~~ Now Function $f: \mathbb{Z}_{>0} \longrightarrow \mathbb{Z}$ s.t. $|f(u) - f(v)| = 1$ if u & v are adjacent.

$\mathbb{Z}_{>0}^2$

graph of $f \longrightarrow$ curve

$\hookrightarrow$ counterpart of a path

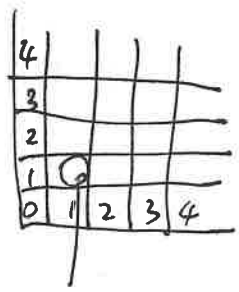
$\rightsquigarrow$ height function

Sample this object. analogously to 1), 2). Know CLT.

2) ~~Originally~~ Now fixed an ~~interval~~ square $\Lambda = [0, N]$ and value of f on ∂N (0 & N)

sampled f on $\Lambda \setminus \partial\Lambda$ uniformly at random.

1) Originally fix a direction, then reveal values of f one by one in this direction randomly.



$\mathbb{Z}_{\geq 0} \to \mathbb{Z}^2_{\geq 0}$

Direction originally $\to$

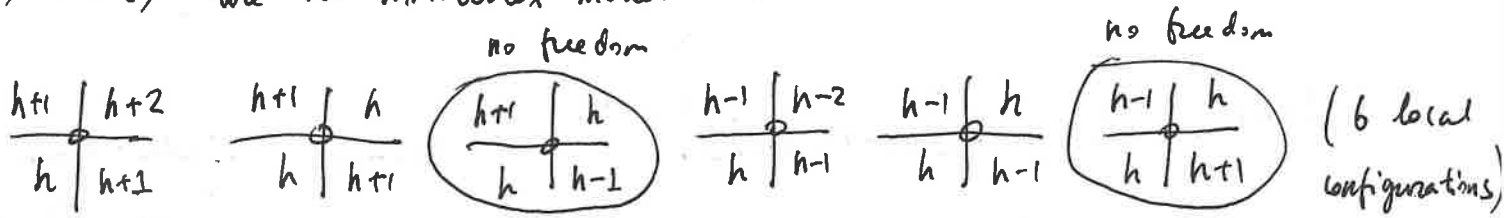
now $\nearrow$

Either 0 or 2 (two choices)

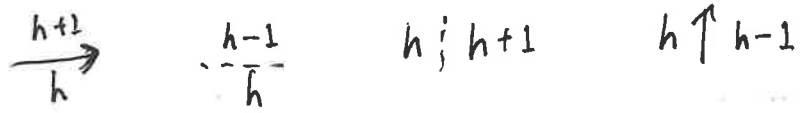
Go up w/ probability p

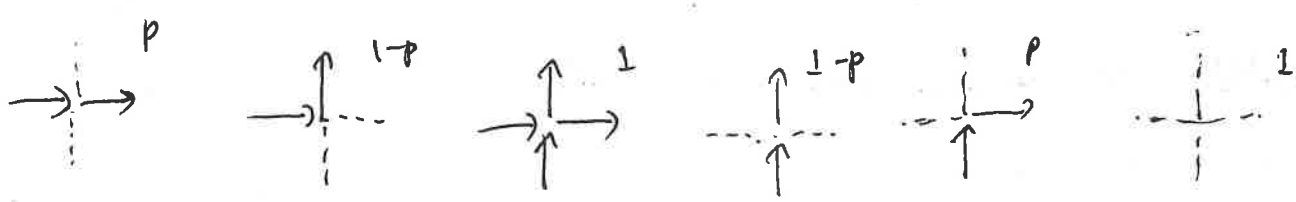
down $1-p$

1) and 2) are the six-vertex model



Convert to arrow configuration:





<u>Bijection</u> between height functions and six-vertex configurations. have arrow conservation

(1) Rule was whenever we have freedom, pick higher choice w/ probability p, and lower choice w/ probability $1-p$.

(2) All weights equal to 1.

$\longrightarrow$ Each configuration has weight 1

$\longrightarrow$ Each configuration $\longleftrightarrow$ height function equally likely

$\longrightarrow$ picking h uniformly at random.

<u>Fluctuation theorems</u>

<u>Theorem</u>. Cases (1) and (2) are very different.

1) $\left(f(N,N) - \mathbb{E}[f(N,N)]\right)N^{-\frac{1}{3}} \longrightarrow$ Tracy-Widom GUE

If $p \neq \frac{1}{2}$ (asymmetric random walk) $\qquad TW_2$

$$\left(f(xN, yN) - \mathbb{E}[f(xN, yN)]\right)N^{-1/3} \longrightarrow \mathcal{H}(x,y)$$ Kardar-Parisi-Zhang (KPZ) fixed point

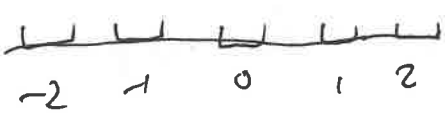
(2) $\left|\operatorname{Var} H\left(\frac{N}{2}, \frac{N}{2}\right)\right| \asymp \log N$

Moreover, $\dfrac{H\left(\frac{N}{2}, \frac{N}{2}\right)}{\sqrt{\log N}} \longrightarrow$ Gaussian

<u>Conj.</u> $H\left(\frac{xN}{2}, \frac{yN}{2}\right) \longrightarrow$ Gaussian free field (GFF)

<u>Lecture 10</u>. Connection between LV and interacting particle systems.

Asymmetric simple exclusion process (ASEP), particles on $\mathbb{Z}$ s.t. ≤ 1 particle at each site

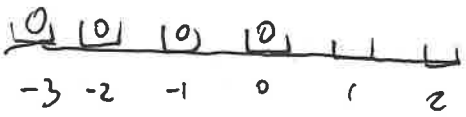


particles try jumping left / right according to exponential clock of rate L / R $\quad 0 \leq L < R$

probability clock rings at time $t \sim L^{-1} e^{-tL}\, dt$

Jump allowed if after it ≤ 1 particle per site, suppressed otherwise.

<u>Thm.</u>



(step initial condition)

Let $N_t(x)$ = # particles right of x at time t.

$$\left[\frac{N_T(xT) - c_x T}{T^{1/3}}\right] \longrightarrow \text{TW distribution}$$

$$\left(\frac{N_T(xT) - c_x T}{T^{1/3}}\right)_{x \in \mathbb{R}} \longrightarrow \text{Airy process}$$

Reminiscent of GV result from before

1	1	b_1	b_2	$1-b_1$	$1-b_2$
		$\frac{q(1-z)}{1-qz}$	$\frac{1-z}{1-qz}$	$\frac{1-q}{1-qz}$	$\frac{z(1-q)}{1-qz}$

bijection between $(q,z) \leftrightarrow (b_1, b_2)$

<u>Rmk</u> Set $b_1 = b_2 = 0$. $\rightsquigarrow$ Each arrow must alternate between right and up steps.

$\rightsquigarrow$ denote the location of the ith arrow by $r_i(t)$ ← vertical coord.

$\rightarrow r_i(t) = r_i(0) + t \rightarrow r_i(t) - t = \Gamma_i(t)$ constant

Now set $b_1 = \varepsilon L$, $b_2 = \varepsilon R$ for small $\varepsilon > 0$

Usually $\gamma_i(t+1) = \gamma_i(t) + 1$ $\rightarrow$ $\Gamma_i(t+1) = \Gamma_i(t)$

Occasionally (w/ probability εL) $\gamma_i(t+1) = \gamma_i(t)$

$\rightarrow \Gamma_i(t+1) = \Gamma_i(t) - 1$ (particle jumps left)

Occasionally (w/ probability $\varepsilon R + O(\varepsilon^2)$) $\uparrow\rightarrow\rightarrow\uparrow$ $\gamma_i(t+1) = \gamma_i(t) + 2$

$\rightarrow \Gamma_i(t+1) = \Gamma_i(t) + 1$ (particle jumps right)

<u>Summary</u>, $\Gamma_i(t)$ constant except it jumps left probability εL,

right —— εR

Set $t = \varepsilon^{-1}\tau$, $\rightsquigarrow$ rare events happen according clocks of rates L and R.

Probability of no left jump in time $[0, \varepsilon^{-1}\tau] = (1-\varepsilon L)^{\varepsilon^{-1}\tau} \longrightarrow e^{-L\tau}$

Look at 6V $(b_1, b_2) = (\varepsilon L, \varepsilon R)$, scale time by ε^{-1}, shift by diagonal $(\gamma \rightarrow \Gamma)$

$\rightarrow$ ASEP

<u>Plan</u>. 1) "Map" 6V to some $\boxed{\text{quantity defined through } F \text{ and } G}$ ← Yang-Baxter

2) interpret in terms of non-intersecting random walks

3) In scaling limit, become non-intersecting Brownian motion

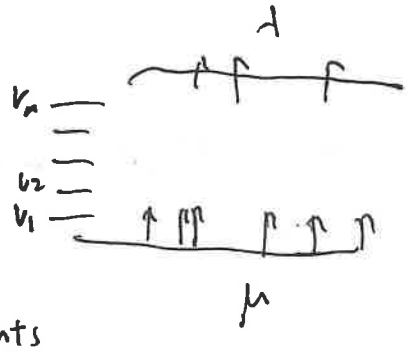
4) Characterisation result for that say TW / Airy process is the only such object.

u				
w	1	u	$1-q^{k+1}$	u
$\tilde{w}$	u^{-1}	1	$u^{-1}(1-q^{k+1})$	1

(A, C)

(B, D)

$P_{\lambda/\mu}(v_1, \dots, v_n) =$

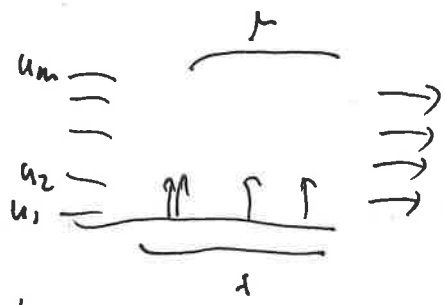


(Hall-Littlewood polynomial)

use w weights

$\leftarrow$ A operators

$Q_{\lambda/\mu}(u_1, \dots, u_m) =$



$\leftarrow$ B operators

use $\tilde{w}$-weights

<u>Rmk</u> P is $s=0$ case of G, Q is a rotation of $s=0$ case of F

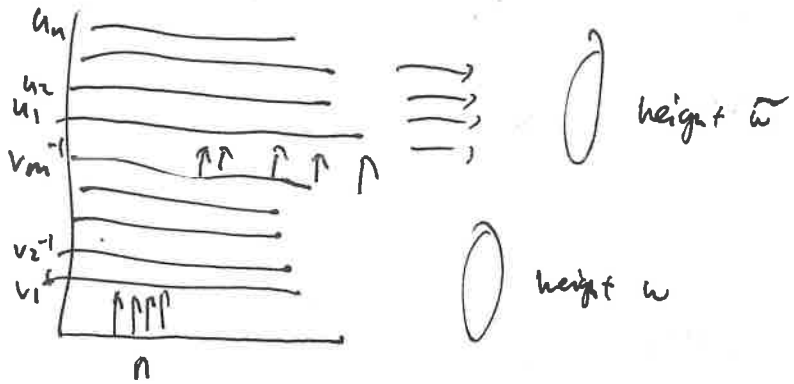
$= F$ after change of variables

$(B \leftrightarrow C)$

<u>Cauchy identity</u>: If $|u_i v_j| < 1$, $\forall i, j$,

$$\sum_{\lambda} Q_\lambda(u_1, u_2, \dots, u_n)\, Q_\lambda(v_1^{-1} v_2^{-1} \dots v_m^{-1}) = \left(\prod_{i,j} \frac{1 - q u_i v_j}{1 - u_i v_j} \right) (q; q)_n$$

Define probability measure.

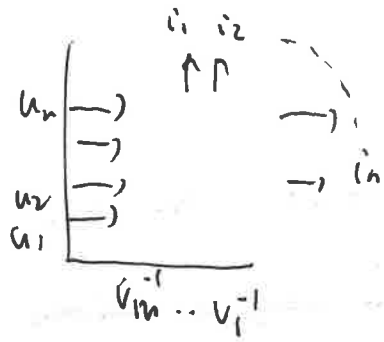


Sample such a vertex model according to weight (model)

$$\cdot\ (q;q)_n^{-1} \prod_{i,j} \frac{1-u_i v_j}{1-q u_i v_j}$$

<u>Cauchy identity</u>: $\longrightarrow$ Probability measure

Plan 1) Match the arrows exiting column 0 in above model to 6V model

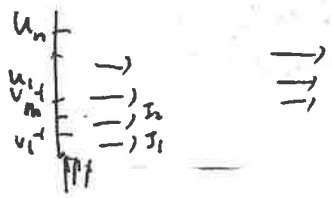


Stochastic six-vertex model w/ the (q,z)-weights from earlier

Recall $z = \dfrac{\text{Horizontal rapidity}}{\text{Vertical rapidity}}$

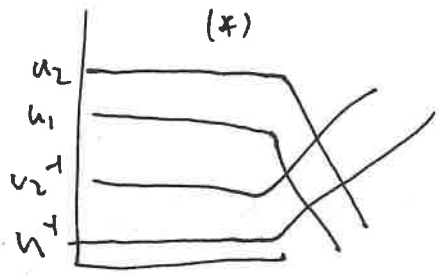
View this as a random set of exits $(i_1, i_2, \dots, i_n)$

Information contain all of $H_t(1)$ = # arrows exit to the right of 1 at time t



(I_j) rows in which arrows exit the leftmost column $I_j \in [1, m+n]$

Theorem. Law of (ij) = Law of (I_j)



Two possibilities for arrow trajectories.

1) All enter cross from left

2) At least one arrow enters cross from bottom.

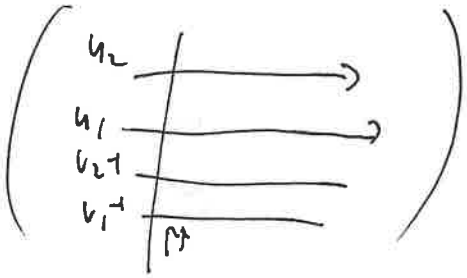
Claim 2) Cannot happen.

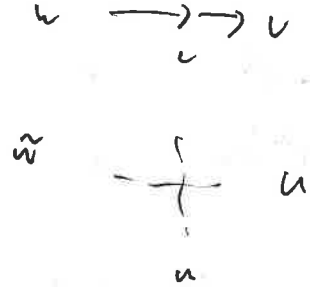
Proof (outline). Suppose some arrow gives right $\mathbb{T}$ spaces in the bottomnost $\rightarrow$ w weight accumulated is V^k

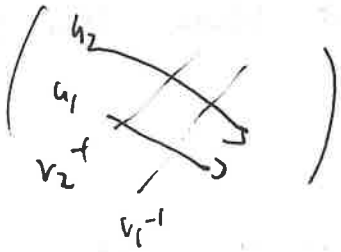
- —

Must have 1)

$\rightarrow$ (*) =





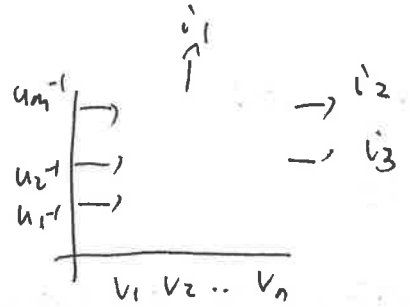


$\mathbb{C}$ $u \longrightarrow \dfrac{1-uv}{1-quv}$

v^{-1}

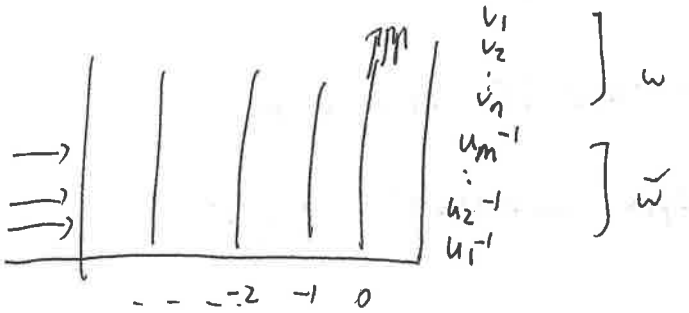
$\prod_{i<j} \dfrac{1-u_iv_j}{1-qu_iv_j}$

Lecture 11 Last time: consider stochastic 6-vertex model on $m \times n$ rectangles w/ domain-wall boundary conditions.

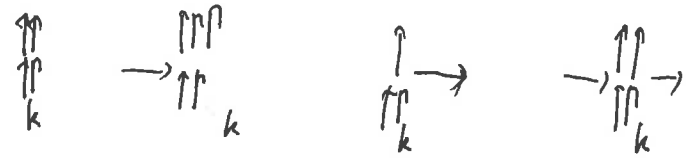


Denote the locations where arrows exit by $i_1 < \cdots < i_m$

Consider HL vertex model on quadrant



(*)



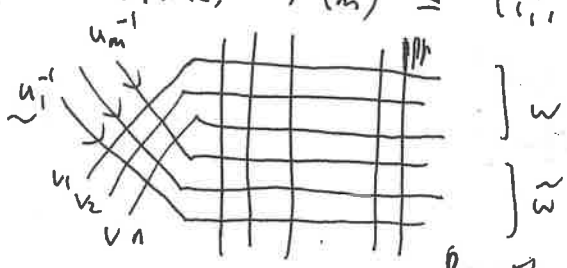
weight type

1	$1-q^k$	v	v	w
u	$(1-q^k)u$	1	1	$\tilde{w}$

Let the rows in which arrows enter rightmost column from top to bottom be $i_1' \leq i_2' \leq \cdots \leq i_m'$.

Matching theorem. $(i_1, i_2, \cdots, i_m) \overset{(d)}{=} (i_1', i_2', \cdots, i_m')$

Proof. (**)



<u>Last time</u>. argued that entering arrows had to go straight through the cross.

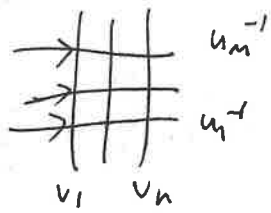
<u>Idea of proof</u>. If an arrow does not go straight through

$\rightsquigarrow$ Arrow moves along v_i row $\frac{L}{m}$ times $\rightsquigarrow (v_i)^{\frac{L}{m}}$

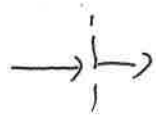
Also, some row u_j^{-1} is not traversed $\frac{L}{m}$ times $\rightsquigarrow (u_j)^{\frac{L}{n}}$

$$\rightsquigarrow \underbrace{(u_j v_i)}_{<1}^{L \min\{\frac{1}{n}, \frac{1}{m}\}} \longrightarrow 0.$$

$(**) = (*)$

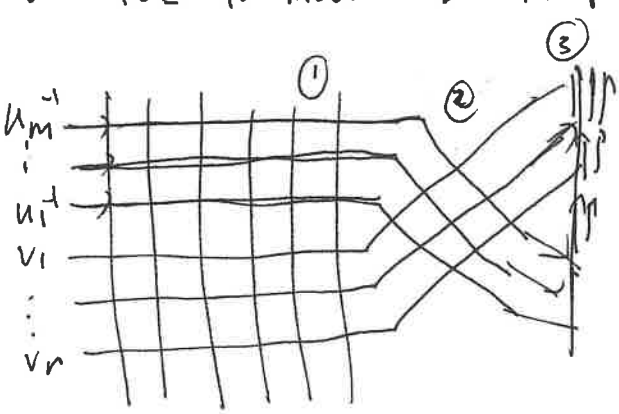


$$= (*) \prod_{i,j} \frac{1-u_i v_j}{1-q u_i v_j}$$



$$\frac{1-z}{1-qz}$$

Use YBE to move cross in front of 0th column.

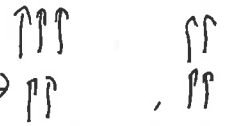


$= (**)$

① $\tilde{w}$ part $\longrightarrow\longrightarrow = 1$

w part $\quad - \uparrow - = 1$

① $\longrightarrow 1$

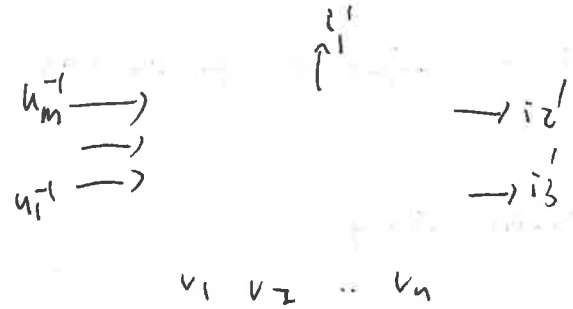
③ $\rightarrow$  ,

$$= \begin{bmatrix} 1-q^{k+1} & (w) \\ (1-q^{k+1})u & (\tilde{w}) \end{bmatrix} \qquad = \begin{bmatrix} 1 & (w) \\ u & (\tilde{w}) \end{bmatrix}$$

product of such weights $\prod_{i=1}^{m} u_i (1-q^i)$

$= (q;q)_m \prod_{i=1}^{m} u_i$

②



$$= \mathbb{P}\left[(i_1 = i_1'), (i_2 = i_2'), \dots, (i_m = i_m') \right]$$

= probability under 6V of $(i_1', \dots, i_m')$

Probability under HL vertex model of $(i_1', \dots, i_m')$ $\sim (*) = (x\,x) \prod_{i,j} \frac{1 - q u_i v_j}{1 - u_i v_j}$

$$= \left((q;q)_m \prod_{i=1}^{m} u_i \prod_{i,j=1} \frac{1 - q u_i v_j}{1 - u_i v_j} \right)$$

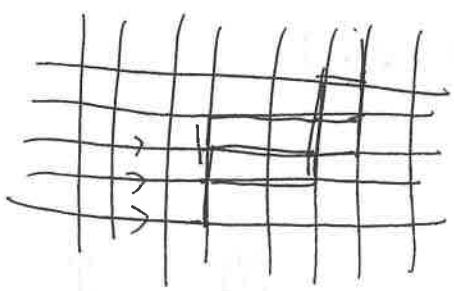
indep. of (i')

$\cdot$ (probability of (i') under 6V)

<u>Line ensembles</u>

$$\mathbb{P}^{HL}((i')) = \mathbb{P}^{6V}((i))$$

<u>Define</u> $L_i(j) = \#$ arrows exiting column i strictly above row j



<u>Properties</u> 1) Since each horizontal edge allows ≤ 1 arrow

$L_i(j) - L_i(j+1) \in \{0, 1\} \longrightarrow L_i$ are Bernoulli paths

2) $L_i(j) \geq L_{i+1}(j) \rightarrow L_i$ are ordered.

If (1), (2) $\longleftrightarrow$ vertex model

Gibbs property

Let us study what happens when we condition (fix) all but a finite rectangle in vertex model.

$$[-k,0]\times[s,t] = R$$

Law of vertex model inside rectangle

$$\mathbb{P}[\text{configuration in } R] = \prod_{v\in R} \overset{(\sim)}{w}(\text{arrow configuration at } v) \qquad (\text{Gibbs property})$$

$$(***)$$

Vertex model

Fix model outside R

① $\longrightarrow$ Fix exiting arrows columns $-k-1, -k-2, \cdots$

② $\longrightarrow$ Fix arrows exiting columns $-1, -2, \cdots, -k$ above t and below s

Line ensemble

① $\longrightarrow$ Fixing $L_{k+1}, L_{k+2}, \cdots$

② $\longrightarrow$ Fixing $L_1, L_2, \cdots, L_k$ outside of $[s-1, t-1]$

Gibbs property for vertex model gives the law of the remaining part $L_1, L_2, \cdots, L_k$

Rmk. $(***)$ only depends on arrows entering and exiting R. (VM)

$\longleftrightarrow$ Law of $(L_1, L_2, \cdots, L_k)\big|_{[s-1,t-1]}$ (LE)

determined by L_{k+1} and end points of $(L_1, L_2, \cdots, L_k)$ on $\{s-1, t-1\}$.

Interested in (i) 6V $\overset{\text{Matching}}{=}$ (i') HL

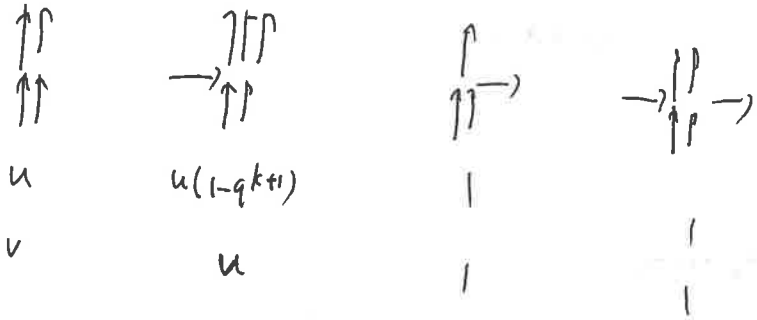
$\overset{\text{def}}{=}$ Arrows exiting -1 column in HL

$\overset{\text{def}}{=}$ L_1.

<u>Closer examination of this law</u>

Take $s, t \in [1, m]$ (only $\tilde{w}$ on VM)

start w/ $q=0$.



$\rightarrow$ power of u every time no arrow exits vertex horizontally.

Weight of R $\sim u^{\boxed{\#\text{ non-horizontal steps in } R}}$

fixed by boundary conditions.

(endpoints of L_i)

~ 1

$\leftrightarrow$ Gibbs property is uniform measure on $(L_1, L_2, \dots)$ satisfying 1) Bernoulli paths
2) Ordered

Condition on L_{k+1}, $(L_i(j))$ $i \in [1,k]$, $j \in \{s-1, t-1\}$

$\rightarrow$ $(L_i(j))$ are Bernoulli random walks conditioned to stay ordered

Speculate on what a limit should look like

1) Bernoulli random walks $\xrightarrow{L_i}$ Brownian bridges B_i

2) Ordered $\longrightarrow$ Ordering $B_1 \geq B_2 \geq \cdots \geq B_{k+1}$

<u>Lecture 12</u> Bernoulli walk bridge $\longrightarrow$ Brownian bridge.

<u>Def'n.</u> Call a family of (random) functions $B_1, B_2, \cdots : \mathbb{R} \to \mathbb{R}$ a <u>Brownian line ensemble</u> if the following holds: for any integer $k \geq 1$, and $s < t$ real conditional on L_{k+1}, and on $(B_1(x), B_2(x), \cdots, B_k(x))$ for $x \in \{s, t\}$, the law of $(B_1(x), B_2(x), \cdots, B_k(x))$ for $x \in [s, t]$ are Brownian bridges starting and ending at $\cdots$, conditional to not intersect, and stay above B_{k+1}.

<u>Characterisation Theorem</u>.

If $B_1(t) = (o(1) - \sigma)t^2$ is <u>approximately a parabola</u> ①, then $(B_1, B_2, \cdots)$ is the <u>Airy line ensemble</u> ③ up to an <u>affine shift</u> ②

① Approximately a parabola: for every $\delta > 0$, $\exists\, C > 1$ s.t.

$$\sup_{t \in \mathbb{R}} \mathbb{P}\left[|B_1(t) + \sigma t^2| > \delta t^2 + C \right] < \delta$$

② Affine shift: $\boxed{X_1, X_2, \cdots}$ is an affine shift of $\boxed{Y_1, Y_2, \cdots}$ if there exist (possibly random but independent of (Y_i)) l and v s.t. $X_i(t) = Y_i(t) + lt + v$.

random functions on $\mathbb{R}$

③ <u>Airy line ensemble</u> Family of functions $A_1, A_2, \dots : \mathbb{R} \to \mathbb{R}$ s.t., denoting A to be the union of the graphs of (A_i), $A = \bigcup_{i=1}^{\infty} \{(t,y) : A_i(t) = y\}$

$$\mathbb{P}\left[\bigcap_{i=1}^{k} (t_i, y_i) \in A\right] = \det\left[K(t_i, y_i, t_j, y_j)\right]_{1 \le i,j \le k} dy_1 dy_2 \dots dy_k$$

Extended Airy kernel
$$K(s, x; t, y) = \begin{cases} \int_0^\infty e^{\lambda(t-s)} Ai(x+\lambda) Ai(y+\lambda)\, d\lambda & (s \ge t) \\ -\int_{-\infty}^{0} e^{\lambda(s-t)} Ai(x+\lambda) Ai(y+\lambda)\, d\lambda & (s < t) \end{cases}$$

$$Ai(x) = \frac{1}{\pi} \int_0^\infty \cos\left(\frac{t^3}{3} + xt\right) dt$$

1) $A_1(0) \sim$ GUE Tracy-Widom distribution

(describe the fluctuation of the highest eigenvalue of a random, Hermitian complex matrix)

2) Ai called Airy process

<u>Rmk</u>

If top curve is not approximately parabolic, then there could exist other (B_i)

There exist Brownian line ensemble, where the top m curves decay linearly

(Airy line ensembles w/ wanderers)

<u>Conjecture</u> 1) Cannot happen that $B_1(t) \ll -t^2$

2) If $B_1(t) \in [-\sigma_1 t^2, C - \sigma_2 t^2]$ w/ high probability, then characterization theorem holds, w/ a (random) scaling.

3) If B_m is approximately a parabola, then $(B_1, B_2, \dots)$ is an Airy line ensemble w/ at most $m-1$ wanderers

Tightness

<u>Theorem</u>. 1) For $q \in [0,1)$, fix x, exist c', c'', c'''

$$\tilde{L}_i(t) = \left(\frac{L_i(xN + ctN^{2/3}) - c'N - c''N^{2/3}}{c'''N^{1/3}} \right)$$

forms a tight family of continuous random curves. any of whose limit points as $N \to \infty$ are Brownian line ensembles

2) $\tilde{L}_1(t) \xrightarrow{N \to \infty} TW + \frac{t^2}{\sqrt{2}}$

<u>Corollary</u> $\tilde{L}_1(t) \to A_1(t)$ Airy process

(Characterisation and tightness)

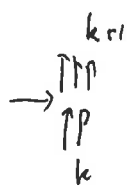
<u>Cor.</u> SGV height function

(<u>Matching</u> and above corollary) $\to A_1$

$H(s,t) = \#$ arrows exiting right of (s,t)

Comments on case $q > 0$

HL vertex weights



$= 1 - q^{k+1}$

Think of k as the distance between L_i and L_{i+1}

in column $-i$

$\rightsquigarrow$ Corresponds to ordered Bernoulli paths that are weighted by factors of $1 - q^{k+1}$ whenever the distance between consecutive paths increases from k to $k+1$.

If theorems are true, then $\hat{L}_1(t) > \hat{L}_2(t) \to L_1(t) > L_2(t) + k N^{1/3}$

$\to k \sim N^{1/3}$

$\to 1 - q^{k+1} \simeq 1 \qquad (q < 1)$

$\to$ should be possible to neglect weights of $1-q^{k+1}$ $\to$ prove tightness.

<u>Remark</u>. 1) $q=0$ SGV model is <u>determinantal</u> so determinant formulas (similar to those defining Airy line ensemble) had already in preprint

2) $q \neq 0$ SGV is (probably) not determinantal.

Much of this works for coloured vertex models

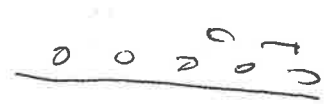
Matching (used YBE) equates behaviour of coloured GV to coloured version HL vertex model. $\xleftrightarrow{\text{bijection}}$ Coloured line ensemble.

Can still use these to prove scaling limits.

<u>Recall</u> GV $\to$ ASEP

Coloured GV $\to$ Multispecies ASEP

Particles jump left / right rates L and R



Now higher labelled particles can jump onto lower labelled ones $\to$ switch.

Lecture 13 Last time: Matching between

1) HL vertex model $\longleftrightarrow$ Stochastic 6V (YBE)

2) HL VM $\longleftrightarrow$ Line ensemble

3) Line ensemble characterisation $\longrightarrow$ Airy process.

Today 1) Briefly mentioning what happens w/ colours

(stochastic 6V $\longrightarrow$ Multispecies ASEP)

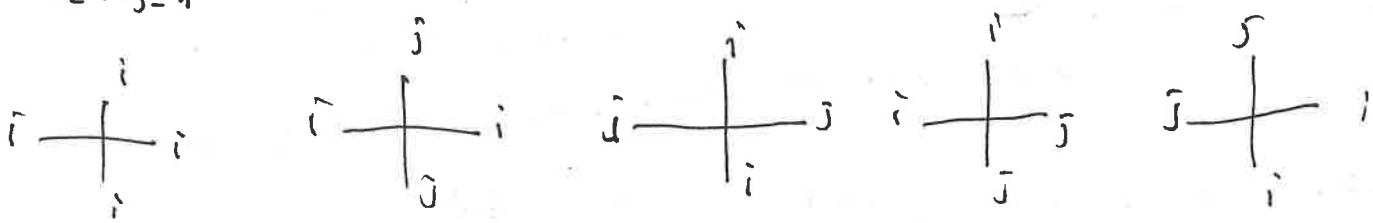
2) AHR Question

3) S6V One point at $q \neq 0$

$\longleftrightarrow$ S6V one point $q=0$

(outline)

$0 \le i < j \le n$

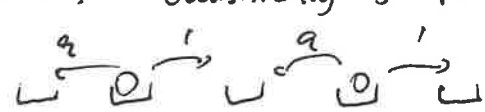


1 $\quad \frac{q(1-z)}{1-qz}$ $\quad \frac{1-z}{1-qz}$ $\quad \frac{1-q}{1-qz}$ $\quad \frac{z(1-q)}{1-qz}$

ASEP limit. $z=1 \rightsquigarrow$ colours zigzag as before, so $r_i(t)$ denote the colour of arrow vertically, exiting site i at time t.

$$\Gamma_i(t) = r_i(t) - t \quad \text{constant in } t$$

$z = 1+\varepsilon$, $\tau = t\varepsilon$, $\varepsilon \to 0$, $\Gamma_i(t)$ usually constant. Occasionally jumps left or right at rates q and 1, respectively.



1)
Jumps of smaller particles onto larger or equal ones will be suppressed.

2) Jumps of larger to smaller happen. ← coloured / multi-species ASEP

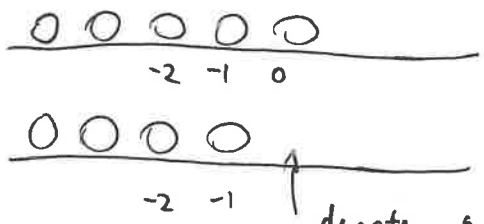
Basic coupling

ASEP (BC)

Each site has two clocks of rates L and R
(left) (right)

When left / right clock rings, the particle on site tries jumping left / right.

Consider ASEPs w/ two different initial data.



Natural way to couple these systems use same clocks.

denote position of colour 1 particle at time t by $X(t)$

Relation between basic coupling and coloured ASEP.

Assume $(\eta_i(t))_{\in\{0,1\}}$ and $(\xi_i(t))_{\in\{0,1\}}$

Denotes whether or not exists particle at site i and time t

$\eta_i(0) \geq \xi_i(0)$, $\forall i$.

Colour all particles in $\xi = \xi \wedge \eta$ (min) colour 2 (heavy)

$\eta / \xi = \eta - \xi$ colour 1 (light) (*)

Claim. Evolution of $(*)$ is a multispecies ASEP.

$\rightsquigarrow$ Multispecies ASEP describes coupled ASEP under BC.

Th. $\lim_{t\to\infty} \frac{X(t)}{t} \to$ uniformly random variable in $[-1,1]$ (almost surely)

$X(t)$ behaves as a line w/ a random slope.

Coloured (ASEP) 6V satisfies YBE.

Colourless case using

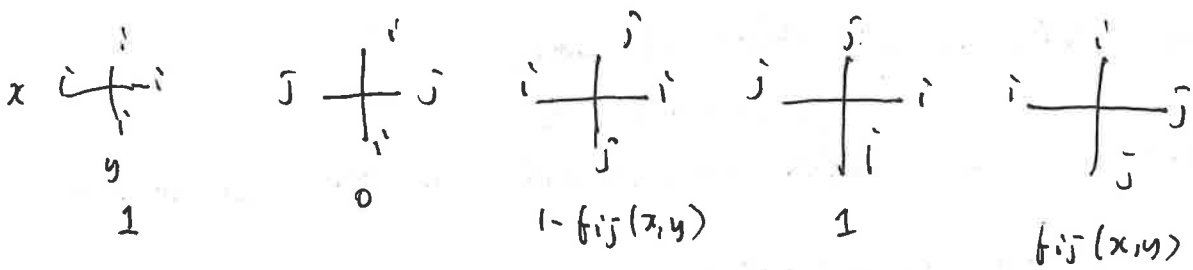
$\square = HL$ weights

($s=0$ fused weights)

$\wr_{q-\frac{n}{2}}$

Arnett – Heinzel – Rittenberg (AHR) model

View this as a coloured model, but where different species have different rates.



1 $\quad$ 0 $\quad$ $1-f_{ij}(x,y)$ $\quad$ 1 $\quad$ $f_{ij}(x,y)$

Two species $0 \le i < j \le 2$ $\qquad$ ("$q=0$")

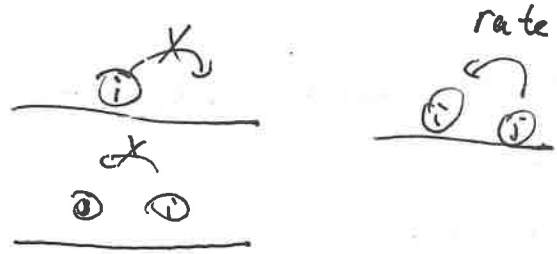
Example $f_{ij}(x,y) = \frac{y}{x} = \beta \rightsquigarrow$ coloured 6V at $q=0$ or ∞ (?)

f_{ij} here independent of i,j

Open. Understand limits of these models for any f that depends on (i,j).
fluctuation

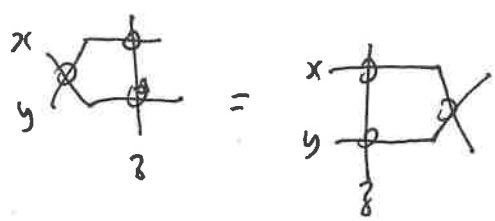
Exclusion limit.

x, y fixed across lattice.



Fact. for certain choices of f_{ij}. $\quad f_{ij}(x,y) = \dfrac{f(x \mid a_i, b_j)}{f(y \mid a_i, b_j)}$

w/ $f(z \mid a, b) = \dfrac{z-a}{z-b}$. weights satisfy YBE

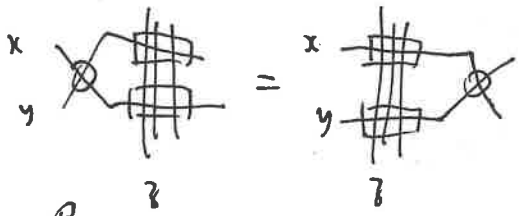


All $\mathcal{O}$ are (**)

Open Understand representation theoretic origin of (**).

Open Want (allow multiple arrows in vertical direction, but at most one in horizontal) ← arbitrarily many

so that

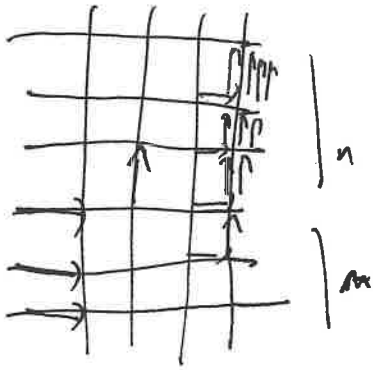


holds for $\mathcal{O}$ as (**)

try solving (w/ computer) for .

Correspondence between general q and $q=0$

HL vertex model



Let # arrows through 0th column at height $m+\frac{1}{2}$

equal $\ell_0 = \ell_0(\lambda) = \#\{0 \in \lambda\}$

weight of HL vertex model

$\sim F_\lambda(u)\, h_\lambda(v)$

$\overset{?}{\sim} P_\lambda(u)\, Q_\lambda(v)$

HL polynomials.

Th. For any $u \in \mathbb{C}$,

$\mathbb{E}\left[(uq^{\ell_0}, q)_\infty^{-1}\right]$ does not depend on q.

Set $u = q^{-z}$ for some z. Recall $q \in [0,1]$.

$$\frac{1}{(q^{\ell_0 - z}; q)_\infty} = \frac{1}{(1-q^{\ell_0-z})(1-q^{\ell_0-z-1})\cdots}$$

$F_z(\ell_0)$

Case 1 $\ell_0 - z \to \infty$. $\longrightarrow q^{\ell_0 - z}, q^{\ell_0-z-1}, \ldots \to 0$

$\to F_z(\ell_0) \to 1$

Case 2 $\ell_0 - z \to -\infty$, $q^{\ell_0 - z} \to \infty$, $F_z(\ell_0) \to O\left(\frac{1}{1+q^{\ell_0-z+1}\cdots}\right) \to 0$

$F_z(\ell_0) \to 0$

If ℓ_0 "spreads" $\longrightarrow \mathbb{E}[F_z(\ell_0)] \approx \mathbb{E}[\mathbb{1}_{z<\ell_0}] = \mathbb{P}[\ell_0 > z]$

<u>Consequence</u> $\mathbb{P}[\ell_0 > z]$ asymptotically does not depend on q

$\longrightarrow$ $q=0$ matches general q.

<u>Proof outline</u>. Property of HL polynomials

$\exists$ difference operators D_r s.t. $D_r P_\lambda = d_r(\lambda) P_\lambda$

$$\rightsquigarrow \sum_{r=0}^{\infty} D_r z^r = D(z)$$

$$D(z) P_\lambda = \Big(\underbrace{\sum_{r=0}^{\infty} z^r d_r(\lambda)}_{d_\lambda(z)} \Big) P_\lambda$$

$$D(z) P_\lambda = d_\lambda(z) P_\lambda$$

$$\sum_\lambda P_\lambda(u) Q_\lambda(v) := \prod_{\substack{u \in U \\ v \in V}} \frac{1-quv}{1-uv} = \Pi.$$

$$\mathbb{E}_\lambda[d_\lambda(z)] = \frac{\sum_\lambda d_\lambda(z) P_\lambda Q_\lambda}{\sum_\lambda P_\lambda Q_\lambda} = \frac{D(z)\Pi}{\Pi}$$

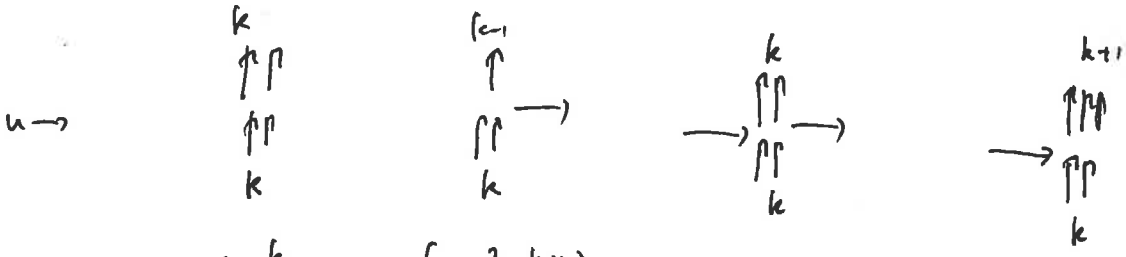
(1)

(2) Evaluate right side and realise it does not depend on q.

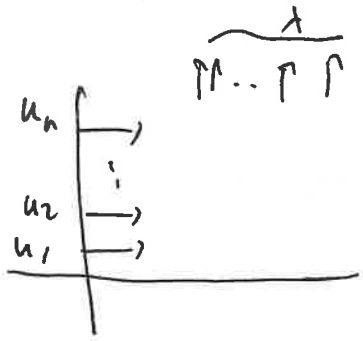
(3) $d_\lambda(z) \sim (z q^{\ell_0(\lambda)}, q)_\infty^{-1}$

Lecture 14 Algebraic Bethe Ansatz

Symmetrization formula for the F- functions



$$\frac{1-sq^k u}{1-su} \qquad \frac{(1-s^2q^{k+1})u}{1-su} \qquad \frac{u-sq^k}{1-su} \qquad \frac{1-q^{k+1}}{1-su}$$



$\lambda = (\lambda_1, \lambda_2, \cdots, \lambda_n)$, $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$

$= F_\lambda(u_1, u_2, \cdots, u_n)$

Arrows exiting at site i = # Times that i appears in λ = $m_i(\lambda)$

Showed 1) Symmetric

2) Branching property (recursion in n)

3) Cauchy identity

Theorem. $$F_\lambda(u_1, u_2, \cdots, u_n) = \frac{(1-q)^n}{\prod_{i=1}^n (1-su_i)} \sum_{\sigma \in S_n} \sigma\left(\prod_{1\leq i<j\leq n} \frac{u_i - qu_j}{u_i - u_j} \prod_{i=1}^n \left(\frac{u_i - s}{1-su_i}\right)^{\lambda_i} \right)$$

σ turns $(i,j) \to (\sigma(i), \sigma(j))$ everywhere

$u_i \to u_{\sigma(i)}$

$\lambda_i \to \lambda_i$

Remark $q = s = 0$.

$$\to \sum_{\sigma \in S_n} \sigma\left(\prod_{1\leq i<j\leq n} \frac{u_i}{u_i - u_j} \prod_{i=1}^n u_i^{\lambda_i} \right)$$

$$= \sum_{\sigma \in S_n} \sigma\left(\prod_{1\leq i<j\leq n} u_i \prod_{i=1}^n u_i^{\lambda_i} \right) \prod_{1\leq i<j\leq n} \frac{1}{u_i - u_j} \quad \mathrm{sgn}(\sigma)$$

$$= \left(\sum_{\sigma \in \mathfrak{S}_n} \sigma \left(\prod_{i=1}^{n} u_i^{\lambda_i + n - i} \right) \operatorname{sgn}(\sigma) \right) \prod_{1 \le i < j \le n} \frac{1}{u_i - u_j}$$

$$= \frac{\det \left(u_j^{\lambda_i + n - i} \right)_{i,j}}{\prod_{i<j} (u_i - u_j)}$$

2) $s = 0$.

$$F_\lambda(u) = (1-q)^n \sum_{\sigma \in \mathfrak{S}_n} \sigma \left(\prod_{i<j} \frac{u_i - q u_j}{u_i - u_j} \prod_{i=1}^{n} u_i^{\lambda_i} \right)$$

def'n of Hall-Littlewood poly.

3) Shift each λ_i by 1, $\lambda_i \to \lambda_i + 1$

$$F_{\lambda + 1^n}(u) = \prod_{i=1}^{n} \frac{u_i - s}{1 - s u_i} F_\lambda(u)$$

$\longrightarrow \longrightarrow$ $\frac{u-s}{1-su}$

<u>Proof of Thm</u>

Recall row operators

A

B

C

D

$$F_\lambda(u_1, u_2, \dots, u_n) = \langle \lambda | C(u_n) C(u_{n-1}) \cdots C(u_1) | \phi \rangle$$

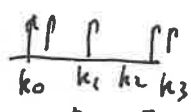
$\langle \lambda |$ represents top boundary for exit data λ

$| \phi \rangle$ represents bottom empty boundary

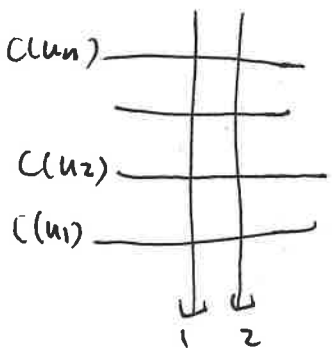
C act on a tensor product of $\bigotimes_{i=0}^{\infty} V_i$,

where $V_i = \operatorname{span}\{ |k\rangle \}_{k \ge 0}$

view V_i as indicating how many arrows leave site i

$|k_0\rangle \otimes |k_1\rangle \otimes \cdots \longrightarrow$ 

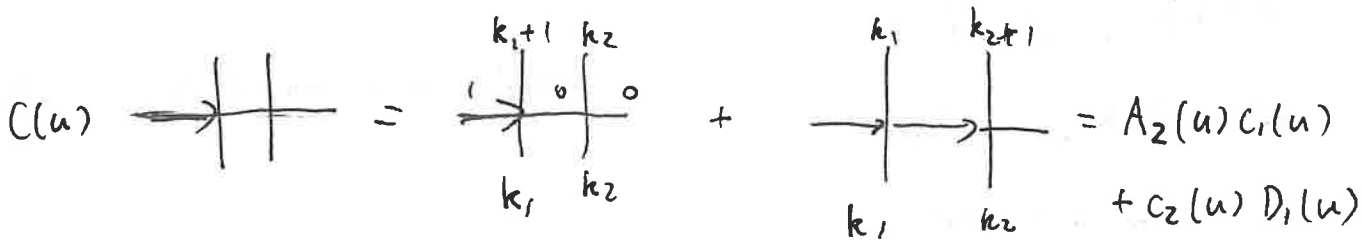
Start w/ two columns.



want to evaluate

$$C(u_1)\,C(u_{n-1})\cdots C(u_1)\,\overset{V_1}{|0\rangle}\otimes\overset{V_2}{|0\rangle}$$

Examine action of C on two columns:



$A_2(u)\,C_1(u)$ acts on V_2 & acts on V_1

$C_2(u)\,D_1(u)$

Recall. $A(u)\,A(v) = A(v)\,A(u)$ (similarly for B, C, D)

$$\left(1-\frac{u}{v}\right)D(v)\,C(u) = \left(1-\frac{qu}{v}\right)C(u)\,D(v) + (q-1)\,C(v)\,D(u)$$

$$\left(1-\frac{u}{v}\right)A(u)\,C(v) = \left(1-\frac{qu}{v}\right)C(v)\,A(u) + (1-q)\frac{u}{v}\,C(u)\,A(v)$$

$$C(u_n)\,C(u_{n-1})\cdots C(u_1)\,|\phi\rangle\otimes|\phi\rangle$$

$$= \left(A_2(u_n)\,C_1(u_n) + C_2(u_n)\,D_1(u_n)\right)\cdots\left(A_2(u_1)\,C_1(u_1) + C_2(u_1)\,D_1(u_1)\right)$$

$$|\phi\rangle\otimes|\phi\rangle$$

$\uparrow_{V_1}$ $\uparrow_{V_2}$

$$\sum \begin{matrix}(\text{polynomial in } C_1, D_1)\,|\phi\rangle \\ (\text{polynomial in } A_2, C_2)\,|\phi\rangle\end{matrix}$$

= Sum of quantities of the form

$$(*) \quad \underbrace{C_1(u_{k_{n-r}}) \cdots C_1(u_{k_1}) D_1(u_{l_r}) \cdots D_1(u_{l_2}) D_1(u_{l_1}) |0\rangle}_{v_1}$$

$$\underbrace{C_2(u_{i_{n-r'}}) \cdots C_2(u_{i_1}) A_2(u_{j_{r'}}) \cdots A_2(u_{j_2}) A_2(u_{j_1}) |0\rangle}_{v_2}$$

$I = (i_1, i_2, \cdots, i_{n-r'})$

$J = (j_1, j_2, \cdots, j_{r'})$

$K = (k_1, k_2, \cdots, k_{n-r})$ Expanding and commutation relations

$L = (l_1, l_2, \cdots l_r)$

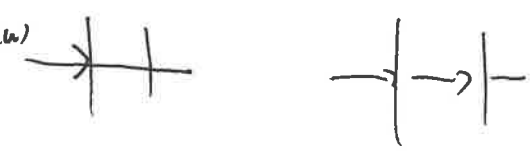
$$F_\lambda(u) = \sum_{I,J,K,L} \text{Coeff}(I,J,K,L) \quad (*)$$

<u>Fact</u>. Coeff does not depend on the order in which the commutation relations are applied.

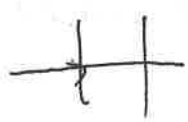
<u>Claim</u>. $\text{Coeff}(I,J,K,L) = 0$ unless $I \cap K$ empty

<u>Proof</u>. Induct on n.

$C(u)$



$C_2(u) D_1(u)$ ← Term 1



$A_2(u) C_1(u)$ ← Term 1

Show that when evaluating $C(u_2) C(u_1) |\phi\rangle \otimes |\phi\rangle$, $1 \notin I \cap K$ for nonzero coeff.

assume. then apply same reasoning to $C(u_1) C(u_2) |\phi\rangle \otimes |\phi\rangle$ $\rightsquigarrow 2 \notin I \cap K$

$$\left(C_2(u_2)\, D_1(u_2) + A_2(u_2)\, C_1(u_2)\right)\left(C_2(u_1)\, D_1(u_1) + A_2(u_1)\, C_1(u_1)\right)$$

Want to show that, upon expanding and then pushing C to left, u_1 is not in a C_2 and a C_1 simultaneously.

<u>Proof</u> in general (two ways)

1) Check other terms by hand (possible but more tedious for general n)

2) Observe that the only way for n_1 to be an argument of C_2 is for $C_2(u_1)\, D_2(u_1)$ to be chosen at the end.

This prevents $C_1(u_1)$ from being chosen $\to$ u_1 cannot appear in a C_1

$\rightsquigarrow$ shows that $1 \notin I \cap K$ $\rightsquigarrow$ $2 \notin I \cap K$ $\to$ $I \cap K$ empty

(Case $n=2$)

General n, $1 \notin I \cap K$.

$\rightsquigarrow$ $m \notin I \cap K$, $\forall m$ $\to$ $I \cap K$ empty.

$$F_\lambda(u) = \sum_{\substack{I,J,K,L \\ I \cap K \text{ empty}}} \text{Coeff}(I,J,K,L) \quad \begin{matrix} C\cdots C\, DD\cdots D\, |0\rangle \\ CC\cdots C\, AA\cdots A\, |0\rangle \end{matrix}$$

$|I| = |L|$, $|J| = |K|$

$I \cap K = \phi$,

$I \cup J$, $K \cup L = \{1,2,\dots,n\}$.

$\rightsquigarrow$ $I = L$, $J = K$.

$$\rightsquigarrow F_\lambda(u) = \sum_{I,J} \text{Coeff}(I,J,J,I) \quad \begin{matrix} \overbrace{C\cdots C}^{J}\ \overbrace{DD\cdots D}^{I}\ |0\rangle \leftarrow u_1 \quad ① \\ \underbrace{CC\cdots C}_{I}\ \underbrace{AA\cdots A}_{J}\ |0\rangle \leftarrow u_2 \quad ② \end{matrix}$$

Plan. Evaluate ①

Evaluate A part of ② left w/ $CC\cdots C|\phi\rangle$ ← V_2

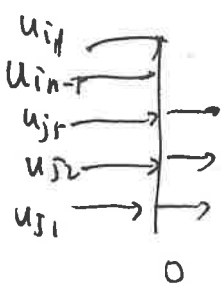
recurse.

$$\overbrace{CC\cdots C}^{I}\ \overbrace{HA\cdots A}^{J}\ |\phi\rangle\ V_2$$

$A|\phi\rangle$ produces $|\phi\rangle$

$$\underbrace{C(u_{j_1})C(u_{j_2})\cdots C(u_{j_r})}_{\text{left in } V_2 \text{ component}}|\phi\rangle$$

$$C(u_{i_r})\,C(u_{i_2})\cdots C(u_{i_{n-r}})\ D(u_{j_1})D(u_{j_2})\cdots D(u_{j_r})\ |0\rangle$$



$$\frac{1-q^{k+1}}{1-su}$$

Each row $\prod_{k=0}^{r}(1-q^{k+1}) = (q;q)_r$

$$(q;q)_r \prod_{m=1}^{n-1}\frac{u_{i_m}-s}{1-su_{i_m}}\prod_{m=1}^{r}\frac{1}{1-su_{i_m}}\,|r\rangle.$$

Lecture 15 Last time: wanted a symmetrization formula

$$F_\lambda(u) = \sum_{\sigma\in S_n}\sigma\left(\prod_{i<j}\frac{u_i-qu_j}{u_i-u_j}\prod_{i=1}^{n}\left(\frac{u_i-s}{1-su_i}\right)^{\lambda_i}\right)$$

$$\sum_k \mathrm{Coeff}(k)\ (q;q)_{|k|}\prod_{l\notin k}(u_l-s)\prod_{k=1}^{n}(1-su_k)^{-1}\prod_{i\notin k}C(u_i)|\phi\rangle$$

Need 1) Evaluate $\mathrm{Coeff}(k)$

2) Recurse

Claim: $\mathrm{Coeff}(k) = \prod_{\substack{j\in k\\ i\notin k}}\frac{u_i-qu_j}{u_i-u_j}$

$$F_\lambda(u) = \sum_{k} \prod_{\substack{i\in k\\ j\notin k}} \frac{u_i - q u_j}{u_i - u_j} (q;q)_{|k|} \prod_{\ell \notin k} (u_\ell - s) \prod_{i=1}^{n} (1-su)^{-1} \prod_{i\notin k} ((u_i)|\phi\rangle$$

$$= \sum_{\substack{k_1\cup k_2\cup\cdots\cup k_m\\ =\{1,2,\dots,n\}}} \prod_{a=1}^{m} \prod_{\substack{i\in k_a\\ j\notin k_a}} \frac{u_i - q u_j}{u_i - u_j} \prod_{a=1}^{m} (q;q)_{|k_a|} \prod_{a=1}^{m} \prod_{\ell\notin k_a} (u_\ell - s) \prod_{a=1}^{m} \prod_{\substack{i\in k_a\cup k_{a+1}\\ \cup\cdots\cup k_m}} (1-su_i)^{-1}$$

Identity: $$\frac{(q;q)_{|k|}}{(1-q)^{|k|}} = \sum_{\sigma\in\mathfrak{S}_k} \sigma\left(\prod_{1\le i<j\le k} \frac{u_i - q u_j}{u_i - u_j}\right) \quad (*)$$

<u>Proof</u> (Outline) 1) Multiply through by $\prod_{i<j}(u_i - u_j)$

2) Realize that the right side is anti-sym. in (u_i)

$\rightarrow$ multiple of $\prod_{i<j}(u_i - u_j)$

3) Degree of right side is (2). $\rightarrow$ $(*)$ constant

4) Evaluate constant : set $u_i = q^i$

$$\sum_{\substack{k_1\cup\cdots\cup k_m\\ =\{1,2,\dots,n\}}} \prod_{a=1}^{m} \quad \cdots\cdots$$

<u>Guessed</u> $$F_\lambda(u) = \sum_{\sigma\in\mathfrak{S}_n} C(\sigma)\, \sigma\left(\prod_{i=1}^{n}\left(\frac{u_i - s}{1-su_i}\right)^{\lambda_i}\right)$$

F_λ can be viewed as eigen functions of a certain row operator $(A+D)$

$$(A+D)\,|\,(c(u_1)\,c(u_2)\cdots c(u_n)|0\rangle)$$

Remains to understand $C(K) = \text{Coeff}(K)$.

Coeff of $\prod_{k \in K} C_1(u_k) \prod_{\ell \notin K} D_1(u_\ell) \prod_{i \notin K} C_2(u) \prod_{j \in K} A_2(u_j)$

$K' = \sigma(K)$ for some permutation σ

original $C(u_n)\, C(u_{n-1}) \cdots C(u_1)$

$\longrightarrow$ $\text{Coeff}(K') = \sigma(\text{Coeff}(K))$ (because F was symmetric)

For any r only need to compute $\text{Coeff}(K)$ for <u>one</u> choice of $K \subset \{1, 2, \dots n\}$ of size r.

$$\left(A_2(u_n) C_1(u_n) + C_2(u_n) D_1(u_n)\right) \cdots \left(A_2(u_1) C_1(u_1) + C_2(u_1) D_1(u_1)\right)$$

Try $K = \{1, 2, \dots, r\}$. Want coeff. of $C_1(u_1)\, C_1(u_2) \cdots C_1(u_r)\, D_1(u_{r+1}) \cdots D_1(u_n)$

$C_2(u_{r+1})\, C_2(u_{r+2}) \cdots C_2(u_n)\, A_2(u_1) \cdots A_2(u_r)$

$r = 1$ $\quad C_1(u_1)$ $\quad$ Rest D_1

$A_2(u_1)\, C_1(u_i) \quad i \in [1, r]$

Rest $C_2(u_1)\, D_1(u_i)$, $i \in [r+1, n]$

$\rightsquigarrow C_2(u_n) \cdots C_2(u_{r+1})\, A_2(u_r) \cdots A_2(u_1)$ ✓ correct order

$D_1(u_n)\, D_1(u_{n-1}) \cdots D_1(u_{r+1})\, C_1(u_r) \cdots C_1(u_i)$ $\leftarrow$ incorrect order

Must apply commutation relations to move Cs left of Ds <u>w/o</u> interchanging (u_i)s at the end $\longrightarrow$ Each $DC \to CD$ cannot interchange spectral param.

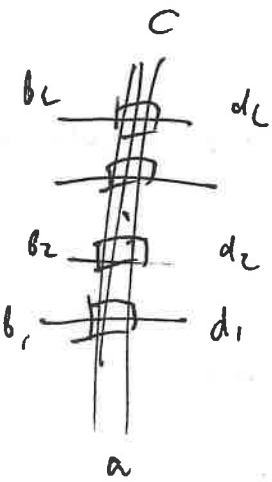
$D(u)\, C(v) = (*)\, C(v)\, D(u) + (**)\, C(u)\, D(v)$. $\quad \text{Coeff}(K) = \text{product}_{\substack{i \le r \\ j > r}} (* \; u_i - u_j)$

Vertex models w/ multiple arrows per horizontal edge.

<u>Recall</u>. We obtained vertex models w/ multiple arrows along vertical edges by starting w/ 6V + and then fusing.

<u>Row fusion</u>

$$W^{L}_{x,y,s}(a,b,c,(d_i)) = \sum_{\substack{(b_i)\in\{0,1\} \\ \sum_{i=1}^{L} b_i = b}} q^{\mathrm{inv}(\overleftarrow{b_i})}$$

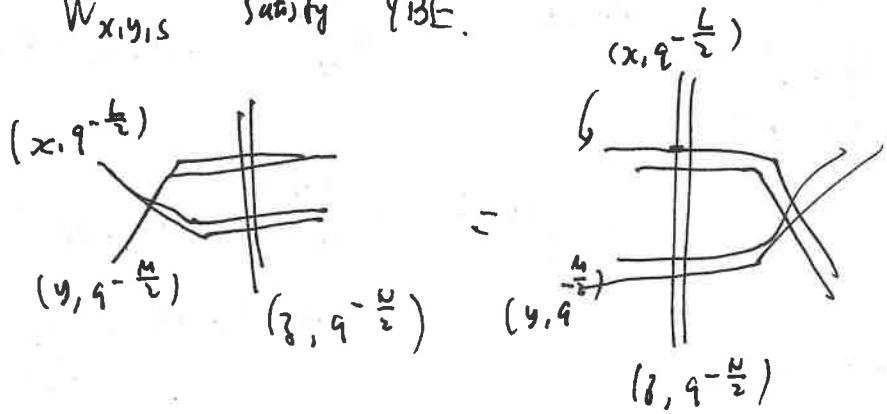


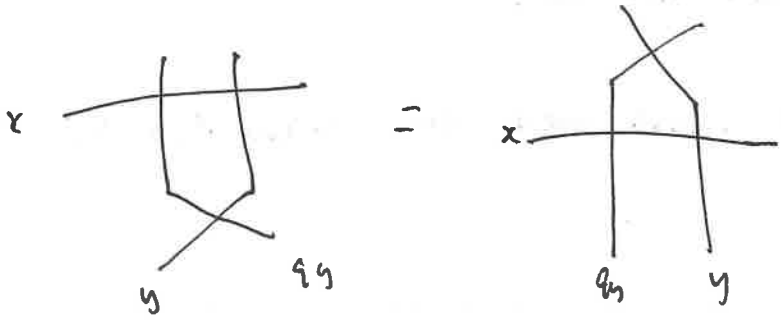
$\square$ spin HL (depend on s)

<u>Prop 1</u> $q^{-\mathrm{inv}((\overleftarrow{d_i}))} W^{L}_{x,y,s}(a,b,c,(d_i))$ $(*)$ does not depend on the choice of (d_i) satisfying $\sum_{i=1}^{L} d_i = d$.

<u>Define</u> $W^{L}_{x,y,s}(a,b,c,d) = (*)$

<u>Prop 2</u> $W^{L}_{x,y,s}$ satisfy YBE.



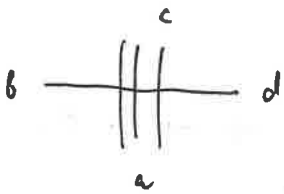


<u>Property</u>

only depends on outputs

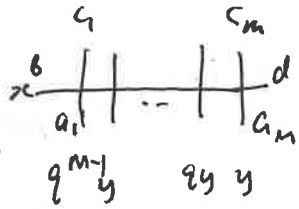
<u>Lecture 16</u> Last time: Second fusion

Before we had spin HL weights



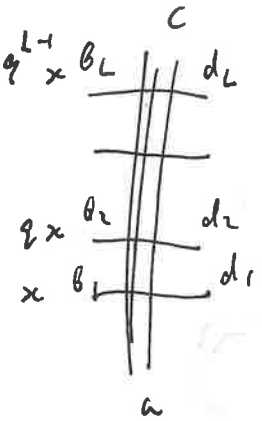
$a, c \geq 0$

$b, d \in \{0, 1\}$

Obtained $\sum_{(a_i)}$ 

$q^{\mathrm{inv}(a)}$

(First fusion)

$\sum_{(b_i)}$ 

$q^{-\mathrm{inv}(\overleftarrow{b})} = W_{\underline{x}}^{(L,s)}(a, b, c, (d_i))$

$\to q^{\mathrm{inv}(\overleftarrow{d})} W_{\underline{x}}^{(L,s)}(a, b, c, (d_i))$

only depends on $\sum_{i=1}^{L} d_i = d$

$W_{\underline{x}}^{(L,s)}(a, b, c, d)$, $a, b, c, d \in \mathbb{Z}_{\geq 0}$

W satisfy YBE

$$ {}_4\varphi_3\left(\begin{matrix} q^{-n}, b, c, d \\ e, f, g \end{matrix} \middle| q, z\right) = \sum_{k=0}^{n} \frac{(q^{-n};q)_k\,(b;q)_k\,(c;q)_k\,(d;q)_k}{(e;q)_k\,(f;q)_k\,(g;q)_k}\, z^k $$

Basic (hyper geometric series)

Regularized BHS

$$ {}_4\overline{\varphi}_3\left(\begin{matrix} q^{-n}, b, c, d \\ e, f, g \end{matrix} \middle| q, z\right) = (e;q)_n\,(f;q)_n\,(g;q)_n\; {}_4\varphi_3\left(\begin{matrix} q^{-n}, b, c, d \\ e, f, g \end{matrix} \middle| q; z\right) $$

<u>Thm</u>

$$ W^{(L,s)}_{\frac{x}{y}}(a,b,c,d) = \mathbb{1}_{a+b=c+d}\, \frac{(-1)^a\, q^{\binom{b}{2}+ab}\, z^a\, s^{b+d-c}\left(\frac{z}{s};q\right)_{d-a}}{(q;q)_c\,(sz;q)_{c+d}\,(q^{L+1-b},q)_{d-b}} $$

$$ \cdot\; {}_4\overline{\varphi}_3\left(\begin{matrix} q^{-c}, q^{-a}, sz\,q^{L}, \frac{qs}{z} \\ s^2, q^{1+d-a}, q^{L+1-c-d} \end{matrix} \middle| q, q\right) $$

<u>Rmk</u> 1) Set $L=1$ $\longrightarrow$ Recover spin HL weights

2) Meromorphic in q^L $\longrightarrow$ can take $r = q^{-\frac{L}{2}} \in \mathbb{C}$.

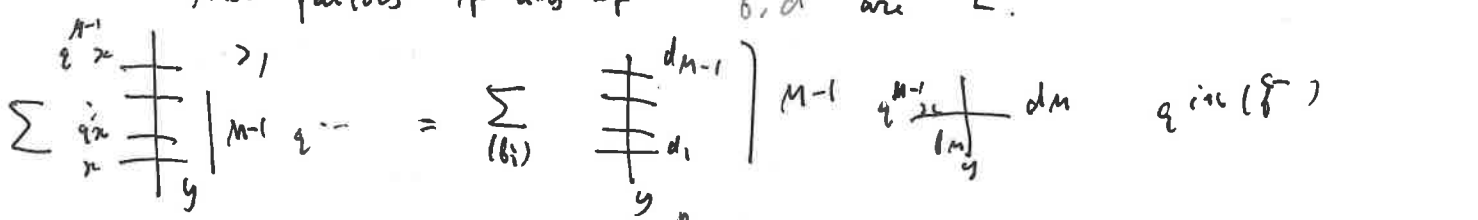
3) Observe that if $c=0$, $\longrightarrow$ ${}_4\varphi_3\left(\begin{matrix} q^{-c}, * & * & * \\ * & * & * \end{matrix} \middle| q, q\right) = 1$

$\longrightarrow$ $W^{(L,s)}_{\frac{x}{y}}(a,b,0,d)$ factors

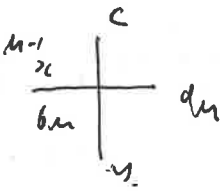
There are transformation formulae that turn $(**)$ into ${}_4\varphi_3\left(\begin{matrix} q^{-a}, q^{-b}, q^{-d} & * & * & * \\ & * & * & * \end{matrix} \middle| q; q\right)$

$\longrightarrow$ $W^{(L,s)}_{\frac{x}{y}}(a,b,c,d)$ factors if <u>any</u> of $a, b, c, d = 0$.

Also factors if any of b, d are L.



$$= \sum_{b_M} q^{\mathrm{inv}((b_M, (b_i)_{i<M})} q^{M-1}$$



Yields a recursion for $W^{(M,S)}_{\frac{x}{y}}(a,b,c,d)$ in M. Can check that

$\overline{{}_4\varphi_3}$ satisfies it.

<u>Issue</u>. Might have troubling guessing that $\overline{{}_4\varphi_3}$ satisfies recursion

<u>Remark</u>. Recursion is also a recursion for

1) $W^{(M,S)}_{\frac{x}{y}}(a,b,c,0)$ and 2) $W^{(M,S)}_{\frac{x}{y}}(a,b,c,M)$

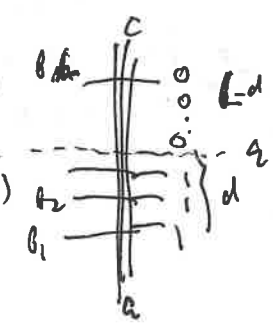
$$W^{(M,S)}_{\frac{x}{y}}(a,b,c,0) = \sum_{a',b',c'} W^{(M-1,S)}_{\frac{x}{y}}(a',b',c',0)\,\mathrm{Coeff}(a',b',c')$$

$$W^{(M,S)}_{\frac{x}{y}}(a,b,c,M) = \sum_{a',b',c'} W^{(M-1,S)}_{\frac{x}{y}}(a',b',c',M-1)\,\mathrm{Coeff}(a',b',c')$$

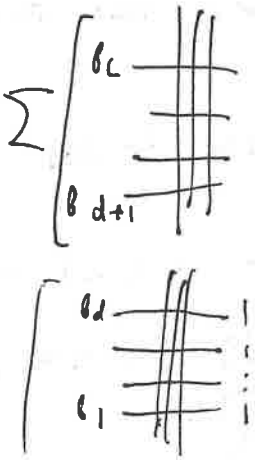
Remark 3) $\longrightarrow$ if $d=0$ or $d=L$, weights should factor

<u>Can guess them</u> and prove them

$$\sum_{(b_i)}$$



$$q^{\mathrm{inv}(b_i)} = \sum$$



$$q^{\mathrm{inv}((\overleftarrow{b_i})_{>d})}\; q^{\mathrm{inv}((\overleftarrow{b_i})_{\le d})}\; q^{\mathrm{inv}((b_i)_{i>d},(b_i)_{i\le d})}$$

$$= \sum \boxed{W^{(L-d,s)}(a',b',c,0)}\ \boxed{W^{(d,s)}(a,b'',a',d)}\ q^{*}$$

Both W are of the previous factored form

$$\to \sum_{a'} \left(\text{factorized quantity}\right) = \square\ {}_4\varphi_3(\cdots)$$

<u>Specialisation of fused weights</u>

Set $z=s$. weights simplify.

$$W^{(L,s)}_{s}(a,b,c,d) = \mathbb{1}_{a+b=c+d}\ \mathbb{1}_{a\geq d}\ (-sq^{L})^{a}\ \frac{(q^{-L};q)_d\,(s^2q^{L};q)_{a-d}}{(s^2;q)_c}$$

$$\cdot\ \frac{(q;q)_c}{(q;q)_d\,(q;q)_{a-d}}$$

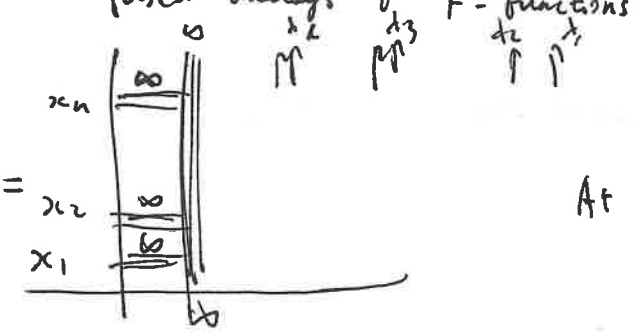
These are weights mentioned in the beginning of the class (spin q-Whittaker wts)

Set $s\to 0$, $q^L\to\infty$ s.t. $sq^L=-\alpha$ fixed $\to \mathbb{1}_{a+b=c+d}\ \mathbb{1}_{a\geq d}\ \alpha^{d}\ \dfrac{(q;q)_c}{(q;q)_d\,(q;q)_{a-d}}$

Further set $q=0$ $\to \mathbb{1}_{a+b=c+d}\ \mathbb{1}_{a\geq d}\ \alpha^{d}$ (Third family of Schur weights from the beginning of course)

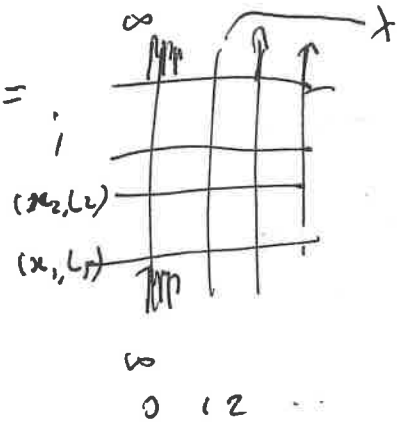
fused analogs of F-functions

$\mathbb{F}_\lambda(x)=$



At site i, have $m_i(\lambda)$ = # parts of λ equal to i

$G_\lambda(x,L) =$



These weights are general $W^{(L,s)}_{\frac{x}{y}}(a,b,c,d)$ weights

L might change across rows

x might change across rows

s fixed

$y = 1$ fixed

Now set each $x_i = s$. weights go from most general $\overline{{}_4\varphi_3}$ fused weights to spin q-Whittaker weights (factored)

$\mathbb{F}_\lambda(s, L) = \mathbb{F}_\lambda(L_1, L_2, \dots, L_n)$ Only functions of (L_i)

<u>Rmk</u> Symmetric functions of (L_i) since the original F was symmetric.

Similarly G symmetric

<u>Lemma</u>. $\mathbb{F}_\lambda(L)$ and $G_\lambda(L)$ are symmetric in (L_i).

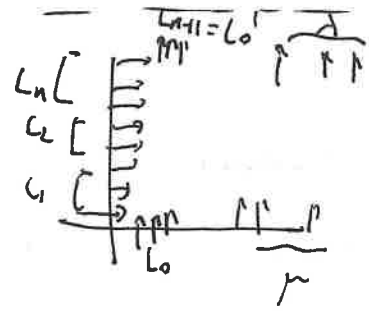
These functions are called spin q-Whittaker polynomials.

$q = 0 \longrightarrow$ Schur polynomials

$q \to 1$ carefully $\longrightarrow$ GL(n) Whittaker functions

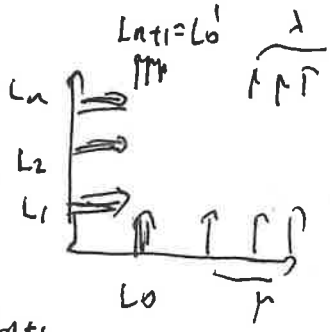
$s=0$, q-Whittaker polynomials are special cases of Macdonald polynomials.

<u>Lecture 17</u> Recall that $\widetilde{\mathbb{F}}^{(s)}_{\lambda|\mu}((x_i),(L_i))$ =



$$= F_{\lambda L_0 L_0' | \mu \cup 1^{L_0}}\left((x_1, qx_1, \dots q^{L_1-1}x_1), (x_2, qx_2, \dots q^{L_2-1}x_2), \dots (x_n, qx_n, \dots q^{L_n-1}x_n)\right)$$

=



φ_3

fused weights

(*) Simplification occurs when we set $x_i = s$ for all i

Fused weights become

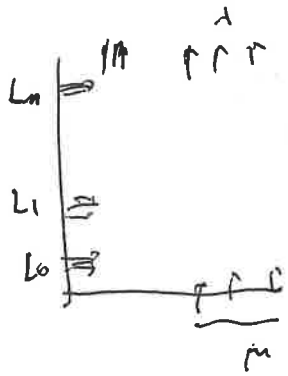
$$(**) \quad W_s^{(L)}(a,b,c,d) = \mathbb{1}_{a+b=c+d}\, \mathbb{1}_{a\geq d}$$

$$(-sq^L)^d \frac{(q,q)_c}{(q,q)_d\,(q,q)_{a-d}} \frac{(q^{-L},q)_d\,(s^{-1}q^L,q)_{a-d}}{(s^2,q)_a}$$

If we set all $x_i = s$ $\rightarrow$ Denote $\widetilde{\mathbb{F}}^{(s)}_{\lambda|\mu}(x,L) = \widetilde{\mathbb{F}}^{(s)}_{\lambda|\mu}(L)$

Partition function of (*) under the weights $w_s^{(L)}$ (**).

To "simulate" L_0 arrows from the bottom, add 0-th row



If μ empty $\rightarrow$ then this recovers $\widetilde{\mathbb{F}}^{(s)}_{\lambda}(L)$

Analogously for $\widetilde{G}^{(s)}_{\lambda|\mu}(L)$.

Properties to hope for:

1) Polynomiality

2) Symmetry

3) Branching recursion

4) Cauchy identity

1) <u>Polynomiality</u>

Set $u = -sq^L$

$\to W_s^{(L)}(a,b,c,d) = \mathbb{1}_{a+b=c+d}\ \mathbb{1}_{a\geq d}$

a) Poly. in u $\to u^d \dfrac{(q;q)_c}{(q;q)_{a-d}(q;q)_d} \dfrac{\left(-\frac{s}{u},q\right)_d (-su,q)_{a-d}}{(s^2;q)_a}$

b) Does not depend on b

Weights are poly. in (u_i)

Finitely many configurations $\longrightarrow$ $\widetilde{\mathbb{F}}^{(s)}_{\lambda/\mu}(L)$ are polynomials in (u_i)

$\widetilde{G}^{(s)}_{\lambda/\mu}(L)$

Write $\mathbb{F}^{(s)}_{\lambda/\mu}(u_1,u_2,\dots,u_n) = \widetilde{\mathbb{F}}^{(s)}_{\lambda/\mu}(L_1,L_2,\dots,L_n,L_0=\infty)$ Similarly for G.

Spin q-Whittaker polynomials ($s=0 \to$ q-Whittaker poly.)

2) <u>Symmetry</u>. $\widetilde{F}^{(s)}_\lambda(L) = F^{(s)}_\lambda\big((s, qs, \dots, q^{L_1-1}s), (s, qs, \dots, q^{L_2-1}s), \dots, (s, qs, \dots, q^{L_n-1}s)\big)$

F sym. $\to$ $\widetilde{\mathbb{F}}$ sym in (L) $\to$ $\mathbb{F}$ sym in (u)

4) <u>Cauchy identity</u> For spin HL polynomials,

$$\sum_\lambda F_\lambda(x_1,x_2,\dots,x_n)\, G^c_\lambda(y_1,y_2,\dots,y_m) = \prod_{i=1}^n \frac{1}{1-sx_i} \prod_{i=1}^n \prod_{j=1}^m \frac{1-qx_iy_j}{1-x_iy_j}$$

$(\Sigma\ (sHL)(sHL))$

Spin q-Whittaker polynomials, there are two Cauchy identities

(a) $\Sigma\ (sqW)(sHL)$, (b) $\Sigma\ (sqW)(sqW)$

$$\overset{sqW}{\sum_\lambda \mathbb{F}_\lambda^{(s)}(u_1,u_2,\dots,u_n)}\ \overset{sHL}{G_\lambda^c(y_1,y_2,\dots,y_m)}$$

$$= \sum_\lambda F_\lambda\big((s,qs,\dots q^{L_1-1}s),\ (s,qs,\dots,q^{L_2-1}s),\dots (s,qs,q^{L_n-1}s)\big)\ G_\lambda^c(y_1,y_2,\dots y_m)$$

$$= \prod_{i=1}^n \frac{1}{(1-s^2)(1-qs^2)\cdots(1-q^{L_i-1}s^2)} \prod_{j=1}^m \prod_{i=1}^n \frac{\cancel{(1-qy_js)(1-qy_jqs)\cdots(1-qy_jq^{L_i-1}s)}}{\cancel{(1-y_js)(1-y_jqs)\cdots(1-y_jq^{L_i-1}s)}}$$

$$= \prod_{i=1}^n \frac{1}{(s^2;q)_{L_i}} \prod_{j=1}^m \prod_{i=1}^n \frac{1-q^{L_i}sy_j}{1-sy_j} = \prod_{i=1}^n \frac{(s^2q^{L_i},q)_\infty}{(s^2,q)_\infty} \prod_{j=1}^m \prod_{i=1}^n \frac{1+u_iy_j}{1-sy_j}$$

$$= \prod_{i=1}^n \frac{(-su_i,q)_\infty}{(s^2,q)_\infty} \prod_{j=1}^m \prod_{i=1}^n \frac{1+u_iy_j}{1-sy_j}$$

$$s=0 \to \sum_\lambda qW_\lambda(u)\,HL_\lambda(y) = \prod_{j=1}^m \prod_{i=1}^n (1+u_iy_j)$$

$$\sum_\lambda \mathbb{F}_\lambda^{(s)}(u)\,G_\lambda^c(v) = \sum_\lambda \mathbb{F}_\lambda^{(s)}(u)\,G_\lambda^{(c)}\big((s,qs,\dots q^{L_1-1}s),\ (s,qs,\dots,q^{L_2-1}s),\ \dots,\ (s,qs,\dots,q^{L_n-1}s)\big)$$

$$= \prod_{i=1}^n \frac{(-su_i,q)_\infty}{(s^2,q)_\infty} \prod_{i=1}^m \prod_{j=1}^m \frac{(1+u_is)(1+qu_is)\cdots(1+q^{L_j-1}u_is)}{(1-s^2)(1-s^2q)\cdots(1-s^2q^{L_j-1})}$$

$$= \prod_{i=1}^n \frac{(-su_i,q)_\infty}{(s^2,q)_\infty} \prod_{i=1}^n \prod_{j=1}^m \frac{(-su_i,q)_\infty}{(u_iv_j,q)_\infty} \frac{(-sv_j,q)_\infty}{(s^2,q)_\infty}$$

Pages